



The Impact of Vision and Leadership on Classroom Technology Deployments in UK Schools

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Declaration of originality

I, Peter Ormerod, declare that I am the sole author of this dissertation, and that the work is a result of my own investigations, except where otherwise stated. All references have been duly cited.

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Author's Background Note

The author is not a teacher, head teacher or professional educator. Having spent several years in the commercial sector supplying classroom technologies around the world, often in very large projects, the author is acutely aware that project outcomes can vary widely. In some cases the impact is rapid and transformational, in others, the effect is more muted and probably doesn't justify the investment. These differences apply not just from country to country, but also from classroom to classroom in the same school.

It's the same technology so that cannot really be the root cause of the variation, and it's almost always too simplistic to just blame the teacher for not enthusiastically embracing change. A hypothesis emerged that there must be something about the process, the culture, the context, the motivation or the vision which impacts the degree of success; ultimately, all of these are strongly connected to leadership.

Hence this choice of topic, to research vision, leadership and change management in relation to such projects, seeking a greater understanding which might be applied to future ventures such that investments might deliver their full potential for educational transformation and advancement of student outcomes.

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My wife has been endlessly understanding and patient, during the many weeks of self-imposed isolation that this study has required. I know of course that it's now payback time! I must also thank the staff at YSJ and RKC for the excellent MALIC course, which has been a perfect fit with my learning needs, and the perfect format to serve as both a catalyst and academic framework for this research.

Abstract

Through a cross-sectional online survey of UK teachers and head teachers, this dissertation explores possible relationships between leadership style and educational outcomes in classroom technology projects.

Amongst an abundance of existing studies investigating the benefits of educational technologies, the importance of effective leadership in such projects is an under-researched field. Questionnaire responses from 1600 participants provided a comprehensive quantitative dataset, which through analysis constructed around a formal research model, investigates four distinct leadership paradigms and assesses the impact on various measures of outcomes including school culture, staff culture, and educational value.

The research is timely, given the growing pressure for schools to continue to improve results under increasingly restrictive budgetary constraints, with technology often being regarded as a means to simultaneously achieve improved educational outcomes and improved operational efficiencies.

The findings of this research indicate that leadership styles which embody Transformational characteristics can bring dramatic additional benefits across multiple outcome measures, as compared to styles which are more Managerial in nature.

For the UK education community this has numerous implications. The emphasis being placed on the transformational leadership model by training bodies appears fully justified, though from the research, these leadership attributes are not yet universally applied so there remains much work still to do.

For international education communities the picture is more complex; this research is somewhat UK centric and would need several adaptations to be fully relevant in other contexts. Yet the fundamental premise remains sound; in order to fully benefit from the considerable investments in classroom technologies, this research demonstrates that differences in leadership style can significantly affect the impact and value of outcomes.

Definitions and Abbreviations

BECTA	British Educational Communications and Technology Agency
ICT	Information and Communication Technologies
IT	Information Technology
DfE	Department for Education
DfES	Department for Education and Skills
EI	Emotional Intelligence
GDP	Gross Domestic Product
MA	Master of Arts
MALIC	Master of Arts in Leading Innovation and Change
OfSTED	Office for Standards in Education
RKC	Robert Kennedy College (Zurich, Switzerland)
SATs	Standard Assessment Tests
UK	United Kingdom
YSJ	University of York St John (York, United Kingdom)

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1 Introduction

For many years and in most national education systems, technology has increasingly formed a significant part of the classroom environment.

Looking back to previous decades, we see that technology of various types has been employed in classrooms since well before the computer era, with innovations such as radio, television, overhead projectors and video being extensively used (Cuban, 1986), though often with little verifiable evidence of improvements in educational outcomes as a result (Schrum & Glassett, 2006).

Today's technology landscape includes classroom computers, interactive whiteboards, learner response systems, laptops, tablets and a range of other high-tech devices. Appropriate choices of technology and their effective use can bring widespread benefits of increased student engagement and understanding, helping learners with differing abilities, styles and cognitive processes, whilst also aiding teacher efficiency (Crook, Harrison, Farrington-Flint, Tomas, & Underwood, 2010).

Yet the opposite case also applies; poor technology choices or badly thought-through implementation and use can be ineffective at best, or highly disruptive and damaging at worst (Cuban, Oversold & Underused, 2001).

Motivations for the increasing ubiquity of technology in classrooms vary; from government level policy imperatives driven by the pressure to succeed in an increasingly globalised economic context (Bottery, 2001), to the local aspirations of educators to '*improve the effectiveness and efficiency of learning*' (Cassidy, 1982).

Some scholars deliberate the role of technology in relation to the very nature of education itself in the 21st century; with the format of the current educational paradigm having been devised to meet the needs of rapidly industrialising societies in previous eras (Albirini, 2007).

Alongside these philosophical perspectives, the more pragmatic amongst us may be drawn to consider the assertion of Cuban (2001) that the '*billions of dollars*' invested in classroom technology have '*yet to produce worthy outcomes*'. There are a wide range of investment options for those seeking continuous improvement of educational provision, so how can we be sure that high levels of investment in technology make best use of limited funding?

In part, some of these issues are absolutely connected to decisions regarding the technologies themselves, where the complexity of strategic level thinking can become confused due to the increasing number of technology choices and unremitting innovation.

However, at their very core, these are really issues of vision and leadership; having a clear purpose and making the right choices, coupled with the ability to orchestrate change to deliver a meaningful, positive and sustainable impact.

The research in this paper assumes as a basic premise that the current educational paradigm will remain dominant and recognisable in form for the foreseeable future; that most school-age students will still attend class and be ordered principally by age; that classes will be ordered by subject around standardised curricula, that most learning will be teacher-led and that the currently prevailing structures and bureaucratic systems of training, assessment and the like will continue *essentially* unchanged.

The aim of this dissertation is to investigate the impact of leadership and vision on the effectiveness of classroom technology projects in the UK.

It is an exploration of how *effective deployment* of technology might achieve a transformation in *educational outcomes*, rather than an investigation of how technology might transform the *educational paradigm* itself.

The research premise is that clarity of vision and effective leadership are critically important in order to gain full benefit from the significant ongoing investments classroom technologies. Through primary research, the question is addressed as to whether classroom technology projects might be perceived to be more or less successful, in the context of the prevailing style of leadership and themes relating to both national and local educational visions. The research follows a predominantly experimental model, testing the null hypothesis that: “*Leadership and Vision are not significant factors in the effectiveness of classroom technology projects in the UK*”.

Testing this hypothesis will incorporate three research objectives:

1. A review of the general and education specific literature on the topics of leadership, vision and classroom technology projects, to identify key variables;
2. A large scale quantitative research project of technology projects in UK schools;
3. To calculate, as effect sizes, the variations from one leadership style to another such that education communities can factor this into their strategic planning.

The primary research is focussed on teachers and principals, rather than at higher levels or administrative functions in the bureaucratic infrastructure.

Section 3 of this paper sets out the detail of the research methodology, the analytical model, and how this is linked to the principal themes arising from the literature review, which investigates a number of aspects of leadership, vision and change. These are heavily researched fields, with a wealth of material spanning different periods, cultures and contexts, studying a range of interconnected themes and overlapping models.

However, specific literature on the research topic of leadership in relation to classroom technology projects appears quite limited; a position acutely summarised by Mike Trucano (World Bank, Senior ICT and Education Policy Specialist) in a personal communication to the author;

“Despite the incredible amount of stuff written on these topics... I find very little of it to be operationally useful. As description or documentation of what's happening? Perhaps. But as useful inputs into decision making? Not so much. The process of translating theory and policy into practice, absent general literature on [educational] change management.... this remains, in my opinion, and in many ways, terra incognita”.

M. Trucano, April 2014

Whilst it might be overly ambitious to hope to comprehensively redress the scarcity of research and fully map this *terra incognita* in the present modest work, the author does hope that the research will be of value in future classroom technology deployments.

2 Literature Review

2.1 The Global Educational Challenge and the Promise of Technology

Almost everything about the world we live in has changed over the last 150 years, yet the fundamental elements of most education systems remain recognisable, even identical, to the models which prevailed in the 19th century (Miller, 2010).

National governments invest significant proportions of GDP to ensure that their citizens have opportunities to acquire the knowledge and skills necessary for productive and fulfilling lives. Under pressure to stay ahead, catch up, reform or become more efficient, education communities in many nations are constantly subjected to initiatives such as President Obama's *Race to the Top* (OECD, 2013) at a cost of some \$4.35 billion in federal grants alone. Internationally, many educational leaders are embarking on their own large scale reform projects.

'Billions have been spent on education reform in the past decade and a half with results in literacy and math, at best, inching forward. This is not value for money, nor is it satisfying for teachers, principals, students and parents'. (Fullan, Hill, & Crevola, 2006).

Whilst these investments and initiatives might span multiple aspects of the educational landscape, during the past 10 to 15 years there has been a particular focus on increasing the role of technology in all facets of education. More than 75% of top education officials around the world feel that technology can play a vitally important and positive role in teaching and learning (CISCO, 2011).

Yet even a cursory internet search for 'failed classroom technology projects' (Google, 2014) uncovers an enormous array of material. Certainly it is possible to find many exemplars of successful projects, such as those highlighted in the RED project (Greaves, Hayes, Wilson, Gielniak, & Peterson, 2012) where there is much to learn for those seeking guidance on delivering transformational change.

However, the wealth of description of failed projects, disillusioned teachers, disruption to learning and sheer waste of money serve as a powerful indicator of the importance of getting it right, and the leadership imperative to find ways of making it so. The spectre of failure appears indifferent to geography, wealth or need; projects in developing countries are as vulnerable as their counterparts in the more privileged nations.

In Pakistan for example, there is a severe education problem; an '*education emergency*', with 40% of the nation's school-age children not in school and huge problems with basic literacy and numeracy (Barber, 2013). Technology solutions here might help address the critical problems faced by developing nations, but additionally, might illuminate novel approaches in other contexts through the notion of reverse innovation (Govindarajan & Trimble, 2012).

Global organisations such as the World Bank invest considerable sums in educational reform, and have teams of experts on hand to advise and assist with project management, often, these specialists express frustration at the poor practices in large scale IT projects (Trucano, 2010).

Yet despite the scale, scope and strategic importance of educational reform using IT, there appears to be relatively little material or research to assist in considering the leadership and change issues specifically for educational IT projects.

'To properly implement technology in our schools, education leaders and policy makers at all levels must have an understanding of technology use from the perspective of school-level leadership' (Greaves et al, 2012).

2.2 Defining Leadership

Leadership is one of the more complex and difficult to grasp aspects of human activity. Repeated attempts to describe and classify the concept have resulted in almost as many definitions as there are researchers in the field (Stogdill, 1974) and we must perhaps conclude that there is no single, exact definition which might apply in all contexts (Pedler, Burgoyne, & Boydell, 2007). Some perspectives are philosophical, where leaders are only considered as such when there are followers who consent to being led, and in one simplistic definition, leadership is just '*...the process of being perceived as a leader*', (Lloyd & Maher, 1993).

It seems that the harder we look for a clear understanding, the more the notion of leadership shows itself to be mysterious and ambiguous (Stewart, 2006).

In the education sector, Leithwood and Jantzi (2006) contend that despite considerable progress in understanding and research during the previous 15 years, the nature of effective school leadership still remains a '*black box*'. Suggesting that although education is a more specific and contextually constrained field than leadership typologies generally, the difficulty in capturing and articulating a precise definition is no less problematic.

2.3 Characteristics of Educational Leaders

A common perception of a 'leader' is of an individual who possesses certain charismatic or heroic qualities; who people will follow largely by choice, rather than obligation. It is a view linked with Thomas Carlyle's early assertion (1849) that '*the history of the world is the biography of great men*'.

Michael Fullan, a recognised international authority on educational reform appears to support aspects of this traditional view of leadership, which he characterises in terms of the personal attributes which provide Hope, Energy and Enthusiasm (Fullan, 2001).

General leadership theory has developed through multiple models since the middle of the 19th century, when Stogdill first challenged trait theory with the observation that people in one situation were not necessarily leaders in other situations (Stogdill, 1948).

Traditionally, most school principals are promoted straight from careers in the classroom, with the title 'head teacher' being an endorsement of the individuals' professional values and competences (Jones, 1999).

Historically, teachers and head teachers have held the respect of society, perhaps almost as parent figures or wider role models. The largely academic and specialised nature of the profession has meant that as a highly educated and trusted community, their internal decision processes on promotion and leadership roles have been relatively insulated from external interference or methods applied in other sectors.

2.4 Models of Educational Leadership

Sergiovani (1979) suggests that there are four distinct models of educational leadership; Scientific Management, Bureaucratic, Collegial and Political (Annex 1, Table A1.1). The approach bases most of its concepts on organisational research in the early 20th century, and although many of the ideas have been displaced by more recent paradigms of organisational leadership, there is considerable merit in these studying four models and relating them to later theory.

In the 1960's there was a growing recognition that managerial activities represented an increasing dimension of the head teachers' role (Jones, 1999); demands which have developed progressively during the 1980's and 1990's, causing dimensions of Bureaucratic and Scientific administration to become increasing components of the traditional Collegial leadership paradigm, characterised in Sergiovani's typology.

Sergiovani's leadership model is somewhat 'old school', in that, whilst useful as a study document, even in 1979 it appears to be an analysis of leadership based on historical, rather than contemporary or innovative models. There is a simplistic approach which would enable these characterisations to map easily onto the Management Grid (Blake & Mouton, 1972), with the Scientific Management and Bureaucratic styles being associated with task, and Collegial and Political styles with team. And whilst published at around the same time, Sergiovani's model incorporates none of the transformational leadership ideas presented by Burns (1978) or the even earlier Institutional leadership ideas of Selznick (1957).

Much of the literature supports the view that traditionally, teaching has been a profession where the members relate to each other on a collegial basis, with promotion to headship often resulting in the head perceiving him/herself as not just the leader of the school, but also the leading exemplar for the school in terms of professional practice (Southworth, 1990). One consequence of this is that head teachers focussed their leadership efforts on what they knew best; effective classroom practice. The phenomena was formally identified in the 1980's, with the classification of the Instructional leadership model, which described the beneficial effects of a high level of direct hands-on leadership involvement in curricula, assessment and high quality classroom practice through teacher observation and mentoring (Muijs, 2010). It was a form of leadership which gained substantial traction and was promoted in many countries, particularly in the US, and most appositely characterised by Hallinger (2003) as *Defining* the schools mission, *Managing* the instructional program, and *Promoting* a positive school-learning climate.

Selznick (1957) contends that there are underlying philosophical components of Institutional Leadership, which form an interesting lens through which to view later models and research into educational leadership.

	Underlying Ethos of Institutional Leadership
Past	Strong sense of past, culture, values, ethos, identity, purpose
Present	Moderate sense of present, managing challenges to preserve identity
Future	Weak sense of future, vision is to preserve long-held views and beliefs, to resist change

Table 2.1 Underlying Ethos of Institutional Leadership,
Selznick, from Washington et al (2008)

However, not all school principals are educational experts, with the time available for instructional activities becoming gradually reduced as school leaders face increased change and growing administrative burdens, often driven by repeated government policy initiatives (Stewart, 2006). New characterisations of the head teacher's role emerged, including Participative Leadership and Transformational Leadership, with the latter being adapted from generalist organisational leadership theory, and both absent from Sergiovani (Table A1.1). Much of this new thinking in educational leadership was driven by the increasing demands for performance and accountability (Stewart, 2006).

Participative Leadership could be considered a *post-heroic* model, in that it represents a shift from a centralised, individualistic or 'lone leader', who demonstrates heroic traits such as authority, courage, control and confidence. In its place is a model of shared leadership, which appears to be an attempt to balance the organisational tensions resulting from the relatively flat academic structures of professional equity, with the hierarchical structures of administration and management. Participative leadership is not something done by one individual to others; rather, it is a group activity that works through and within relationships (Bennet, Wise, Woods, & Harvey, 2003).

Whilst this model might satisfy some of the traditional collegiate aspects of educational organisations, observers might question who is actually in control, who has authority, where does the buck stop? Importantly, who is setting the direction, owns the vision and provides authentic and authoritative organisational leadership?

2.5 Transformational Leadership

The UK government established the National Centre for School Leadership in 2002, specifically for the purpose of developing transformational leadership methodologies (Currie & Lockett, 2007).

Transformational leadership has four principal elements; charisma, inspiration, individualised consideration and intellectual stimulation. Burns (1978) is often credited with introducing the idea of Transformational Leadership arising from his studies of political leadership, which "*appeals to the moral values of followers in an attempt to raise their consciousness about ethical issues*", Yukl (2013).

Burns contrasted transformational leadership with transactional leadership which “*motivates followers by appealing to their self-interest*”.

Inherent in these apparently opposing models is the fundamental issue of motivation; that transformational leadership appeals to high-order motivations of self-actualisation, whilst transactional leadership addresses more basic rewards (Maslow, 1943).

Critics of the transformational leadership model suggest that the appropriate style is context specific. This view is articulated most clearly by Bass and Avolio (1994) who through their distinct definitions of transactional and transformational styles, provide an insight into how leaders might on the one hand adopt a polarised approach, yet on the other might blend both styles according to context.

Key Factor		Behaviours
Transformational Behaviours	Idealised Influence	Leaders act as role models, are admired, respected and trusted, consider the needs of others over their own; are consistent in their behaviours; share risks with others and conduct themselves ethically.
	Inspirational Motivation	Leaders motivate and inspire others by providing meaning and challenge; they rouse team spirit; are enthusiastic and optimistic; communicate expectations and demonstrate commitment to shared visions.
	Intellectual Stimulation	Leaders encourage innovation and creativity through questioning assumptions and reframing problems. They avoid public criticism.
	Individualised Consideration	Leaders attend to individual needs for achievement and growth, engage in coaching and mentoring, create new learning opportunities, value diversity and avoid close supervision.
Transactional Behaviours	Contingent Rewards	Leaders provide rewards on condition that followers conform with performance targets.
	Management by Exception	Leaders take action when task related activity is not going according to plan.

Table 2.2 – Transformational and Transactional Leadership Behaviours.

Within education, transformational leadership could be considered as a hybrid construct, embodying aspects of other leadership forms, such as Moral leadership, which focusses on the values and ethics of the leaders themselves (Hodgkinson, 1991), and Participative leadership, where through shared decision making, staff are bonded together in pursuit of a shared goal (Bush, 2007).

One of the interesting aspects of the transformational leadership model in education, is the implication that key functions which are at the core of other leadership paradigms, such as Bureaucratic and Instructional, *must* be delegated to other players in the schools management team.

Transformational leadership would appear to be a highly appropriate model for schools in the 21st century, particularly in respect of vision and transformational change relating to technology projects. The potential for transactional components to play a role within the framework of transformational leadership would appear to support the increasing need for schools to be rewarded according to league tables and other simplistic measures of performance (Bass & Avolio, 1994). Yet despite this, much of the literature which advocates the value of transformational leadership in education appears strong on rhetoric, but weak on evidence (Currie & Lockett, 2007). It may not be the panacea that some policy makers believe it to be.

2.6 Leadership and Technology Integration

Technology is becoming increasingly ubiquitous, and permeates almost aspects of 21st century life. Yet in the classroom, some express the view that students are '*overwired and undertaught*' (Pool, 1997), criticizing the excessive focus on '*glitzy toys*' (Bronner, 1997) when many schools are failing to provide students with adequate competences in literacy and numeracy.

One of the frequent criticisms is not of the technology itself, but of the way that it is integrated into the classroom environment; that technology will be ineffective if it is simply *placed* in classrooms and must be integrated in a way which supports new instructional and learning experiences (Earle, 2002); integration is not about the amount or type of technology, but in how and why it is used.

Bernauer (1995), commenting on instances of successful technology deployments makes the point clearly; '*it is not technology per se that has resulted in improved student outcomes, but rather, how the technology was used and integrated into the instructional processes*'.

Some go further, asserting that technology is only one component of what must be a broader range of innovation and change initiatives in pedagogy, curriculum, assessment and school organization (Dede, 1997). Ultimately, successful integration is a function of effective change management.

2.7 Leadership and Change Management

Major change can be problematic for any organisation and a challenge for any leader. The first step is a general acceptance of the need for change, which in large, complex organisations is far from straightforward and especially where that change might be regarded as presenting a discontinuous challenge rather than incremental improvement (Nadler, Shaw, & Walton, 1995).

In Lewin's basic three stage change model (1951), accepting and agreeing the need for change, or 'unfreezing' is a critical first step to overcome the inertia of the status quo. The change itself can be a period of confusion and uncertainty, where many will seek to hold onto the old ways and reject the challenge and value of the new. Lewin proposes a force field analysis tool, in which the *driving forces* and *resisting forces* which might impact on the proposed change might be evaluated. Eventually, the aim is for the new model to become accepted and embedded.

Kotter (1996) expands Lewin's three stages into a more sophisticated model of eight distinct phases. Fundamental to Kotter's approach is the importance given to developing a clear change vision, which can be articulated to, and understood by, all stakeholders.

The vision establishes a need for urgent action and ensures continuity and alignment of purpose and action. Kotter's approach is particularly appropriate for top-down projects which, in the context of this paper, might result from a school leaders 'directive' or perhaps even a government policy level initiative.

However, a weakness of Kotter's model is that there are no specific mechanisms for managing conflicts and resistance to change. The assumption appears to be that if the vision is attractive and clearly articulated, any potential for resistance will be neutralised and there will be universal 'buy-in'.

This is especially relevant in the context of education, where individual teachers are generally intelligent, somewhat autonomous, and (in certain cases) hold on to that traditional perspective of their institution, and their role within it. Pelgrum (Obstacles to the integration of ICT in education: Results from a worldwide educational assessment, 2001), has summarised what might represent restraining forces according to specific dimensions, through a comprehensive analysis of the literature.

Resisting Forces in Education Technology Projects		
	Force(s)	Source(s)
1	Access to hardware and software as well as funding	Hope, 1997; Lan, 2000, Legget & Persichitte, 1998; Lumey & Bailey, 1993
2	Time for planning, personal exploration, online access and skill development	Duffield, 1997; Hope, 1997; Lan, 2000; Sheingold & Hadley, 1990
3	Technical and administrative support and resources	Legget & Persichitte, 1998; Schrum, 1995
4	Training and expertise	Cafolla & Knee, 1995; Hope, 1997; Shelton & Jones, 1996
5	Resistance, passivity, school cultures and traditions of teaching	Beachan, 1994; Cafolla & Knee, 1995; Cohen, 1987; Cuban, 1986; Ertmer, 1999; Hope, 1997; Lumely & Bailey, 1993
6	Vision and leadership	Cafolla & Knee, 1995; Ely, 1995; Hope, 1997; Lumely & Bailey, 1993;
7	Support for integration of technologies into instruction and the curriculum	Cuban, 1986; Hancock & Betts, 1994

Fig 2.3 Resisting Forces in Education Technology Projects (Pelgrum, 2001)

Some of the underlying research for this summary was conducted over 15-20 years ago and the dramatic increases in the scale and scope of educational technology during the last 5 years suggest that an updated view is needed. However, its structure is essentially sound, and it remains a useful template for those planning technology projects.

Amongst the few specific references for planning the change processes of classroom technology deployments is the ICT-in-Education Toolkit (Haddad, 2014) which defines a sequential process for project design and execution (Figure 2.4).

	Stage
1	Map the National, Technological, and Educational Situation
2	Formulate and Assess ICT-Enhanced programs
3	Plan for Physical and Human Requirements
4	Plan for ICT-Enhanced Content
5	Generate Program Costs
6	Create a Master Plan
7	Monitor Implementation, Effectiveness, and Impact

Figure 2.4 ICT-in-Education Toolkit (Haddad, 2014).

This an entirely logical planning sequence, which is claimed to have been used (or considered) in around 20 countries. However, whilst the *human requirements* are referenced as part of phase 3, the overall model falls well short of addressing the full implications of vision, leadership, culture and change management issues.

Holbeche (2006) reminds us that; “*managing change is not about managing change; it’s about managing people*”, and from Kotter and others, we know that fundamental to the process of successful change is the vision of change itself.

2.8 Developing a School Technology Vision

If we accept the premise that a successful educational technology project requires a clear vision and effective leadership to align the organisation around the vision and drive towards it, then we must ask, from where does such a vision originate? Interestingly, much of the literature skims over this crucial topic, with ‘vision’ being referred to as a sub-topic or a component of a more specific subject such as transformational leadership or change management.

It appears taken for granted that somehow, a vision will be defined or will be found; from national policy, by stakeholders such as schools governors, or from the school principal as part of the leadership function.

The literature, it seems, is a surprisingly weak source of guidance on this most fundamental of subjects. A non-scientific internet exploration by the author of the websites of some 20 UK schools revealed that around 70% of these had an easy to find *mission statement*, which, for the purposes of this review, we will accept as a short-hand version of the schools strategic vision. However, in around 60% of these the wording is predominantly present tense; which is understandable, in that schools no doubt seek to present a positive impression of what they aim to provide for *today’s* students, rather than what they aspire to be for *tomorrow’s*.

However, thinking back to the underlying school philosophical components described by Selznik (Table 2.1), there is an implied acceptance in many of the mission statements that the school principals’ role is to maintain the existing ethos and status quo, rather than proactively seek continuous improvement opportunities and strive to achieve them.

Guidance for school leadership on the development of a *technology vision* generally appears superficial and lacking in visionary perspectives. Kim Cofino (2012) suggests a simple four stage model; Brainstorm, Organise, Revise, Statements. It is a ‘vision by committee’ approach which might fit well in the context of Participative leadership.

The role of leadership in this model is defined simply as being to “*provide resources and cultivate a supportive and collaborative teaching and learning environment*”, implying that the overall vision is determined collectively, with the leaders role being that of a facilitator.

Ferriter (2011) connects school leaders to the issue of vision by posing the question “*If I asked you to explain the rationale behind the technology choices that your school is making, could you do it?*” Sadly, the model he proposes to help leaders develop a response to the question is a simplistic hand-out which invites participants to consider digital opportunities, digital portfolios, digital warehouses and digital survey and response tools.

The hand-out begins well by suggesting that readers consider their school mission statement, but leaps unhelpfully from there directly to these detailed technical matters, with no apparent consideration of an educational vision and a clearly articulated purpose for the technology, or how it might be properly integrated to fulfil educational goals.

This weak ‘vision’ guidance to educators appears commonplace, and particularly disappointing in respect of leadership for technology projects. A typical example being the 2010 BECTA report offering a wide ranging analysis of classroom technologies (Crook C. , Harrison, Farrington-Flint, Tomás, & Underwood), which whilst mentioning that successful projects tended to be those which had some form of vision, omits any further guidance on the topic.

Looking at literature outside the education sector we find a slightly improved position, with the exemplar work of Nanus (Visionary Leadership, 1992) stating that ‘*there is no more powerful engine driving an organization toward excellence and long-range success than an attractive, worthwhile, and achievable vision of the future, widely shared*’. The themes developed by Nanus on defining and implementing a vision are adapted for education by Leithwood et al (1997). They stress the importance of providing staff with a sense of purpose, energising colleagues through espousing what the vision will accomplish, being clear on what the vision means in terms of practical, daily actions, and using all available opportunities to communicate the schools vision to staff, students, parents, and other members of the school community.

This model is of course applicable at the macro level overall school development visions, yet it is equally valid for the development and articulation of a school technology vision. From the literature, it is clear that not all leadership styles are equal in terms of their understanding and relationship with the concept of vision.

2.9 Leadership in Education; **The Real World**

The models and characteristic of leadership explored so far have been considered mainly as discrete academic concepts, and the question of which paradigms can actually be observed, in practice, in real schools in the UK has not been addressed.

Currie and Lockett (2007) researched this precise topic by carefully analysing OfSTED reports for a large number of schools, defining a range of variables and a robust coding scheme based on a textual analysis of the reports, then extrapolating from this research an analysis of the predominant leadership style within the school.

Leadership Style	Frequency
Professional	86
Transformational	26
Managerial	12
Moral	10
Contingent	5
Participative	4
No clear leadership style, or unsatisfactory leadership	10

Table 2.4 - Characterisations of Educational Leadership (Currie & Lockett, 2007)

They concluded that individualistic rather than shared leadership styles were dominant, and that Moral and Participative approaches were less frequent. The research suggests that Professional (bureaucratic) leadership is the prevalent form, which embodies mission setting and promoting school climate, as well as instructional elements and managing the professional programme (Hallinger & Murphy, 1985).

We might conclude that in the real-world, hybrid models of educational leadership are common, blending the traits and skills of the leader, the cultural and social contexts of the school, along with the demands and motivations of stakeholders and policy makers.

2.10 Research Directions Arising From The Literature Review

The literature review illuminates some of the challenges faced by the education community in the 21st century, and explores issues related to effective utilisation of technology. Certainly, the potential for transformational effects has been demonstrated, though realising full value (both economic and educational) is often difficult and certainly not an inevitable consequence of technology deployments. Vision, leadership and integration would appear to be essential elements, if not always fully understood or applied concepts.

Understanding and defining leadership in schools has progressed considerably during the last century, with several models being observed and characterised. These would appear to reflect many of the leadership paradigms found in the general literature, though with aspects of certain typologies such as institutional and collegiate being more prevalent, or more likely, that some leadership styles have evolved to become more *specialised* as a consequence of the educational context in which they developed.

The work of Leithwood et al (2012), summarising 121 educational journal articles, identified 20 leadership concepts which they consolidated into six distinct paradigms. For the research in this dissertation, drawing heavily on Leithwood's work, the number of leadership paradigms to be studied has been reduced to four, using the methodology set out in Annex 2.

These four leadership are mapped across the two main dimensions of interest; whether leadership is centralised, decentralised, etc., and whether the vision is progressive or static, advocated or assumed.

Distribution of Leadership	High	PARTICIPATIVE	TRANSFORMATIONAL
	Low	MANAGERIAL	INSTRUCTIONAL
		Low	High
		Advocacy of Vision	

Table 2.5 - Leadership paradigms mapped onto research dimensions

The paucity of research in respect of leadership in classroom technology projects forms the basic motivation for this dissertation, with primary research being specific designed to explore these largely undocumented relationships.

3 Research Methodology

The objective of the research is to understand whether there are any significant positive relationships between leadership style, vision and the effective use of classroom technologies in improving educational outcomes.

3.1 Research Philosophy

The ‘research onion’ (Saunders, Lewis, & Thornhill, 2012, p. 128) was a source of useful guidance to inform the overall design philosophy and methodology.

From the very start of the project, it was felt important to ‘begin with the end in mind’ (Covey, 2004), which forced an early consideration of possible research models and potential interpretations and conclusions (Crotty, 1998). Equally importantly, it encouraged reflection of the philosophical aspects of the research and how pre-existing assumptions might support the research whilst not influencing the impartiality of the outcomes (Johnson & Clark, 2006).

Ontology	Epistemology
More Objective than Subjective	More Positivist than Interpretivist
More Realist than Postmodernist	More Empiricist than Constructivist

Table 3.1 – The authors’ simplistic self-assessment of research philosophy

Whilst Table 3.1 may not represent a comprehensive statement of the authors research values (Heron, 1996), it is a useful perspective from which to qualify a broad understanding of personal characteristics with respect to research. It suggests a preference for a structured methodology, for gathering hard data which can be analysed and quantified, and which might subsequently lend itself to replication (Gill & Johnson, 2010).

These philosophical dimensions represent a spectrum, rather than binary, polar opposites. So as a pragmatist, the author felt drawn to a hybrid research model; essentially Experimental, utilising Survey and Archival Research as associated components (Saunders, Lewis, & Thornhill, 2012).

The research model is essentially ‘mono method quantitative’, with a large scale survey at its core. The analysis model is predominantly deductive, though with elements of inductive reasoning in developing outcomes and actionable frameworks arising from the research.

3.2 Research Framework

At an early stage in the planning, and based on guidance in texts such as Saunders et al, it was felt that a large scale online survey of teachers and head teachers would yield the most useful data and perspectives.

The scale of the target population (using a database of 22,000 UK school email addresses, with an anticipated response rate of around 2-3%) meant that a large dataset of responses might be expected. The research is cross-sectional in nature, capturing opinions at a fixed point in time.

The work of Leithwood and Levin (2005), researching educational leadership effectiveness, provides numerous useful insights and models, from which an amended version of their research framework (Figure 3.1) has been derived. It is based on a generic research model, and provides a conceptual structure through which the themes of interest from the literature review can be classified, thus forming the top level basis for the organisation of the research. Study of Leithwood and Levin's methodology assisted in categorising topics according to their role in the framework; as Antecedents, Independent Variables, Moderating Variables, Dependent Variables, or Mediating Variables.

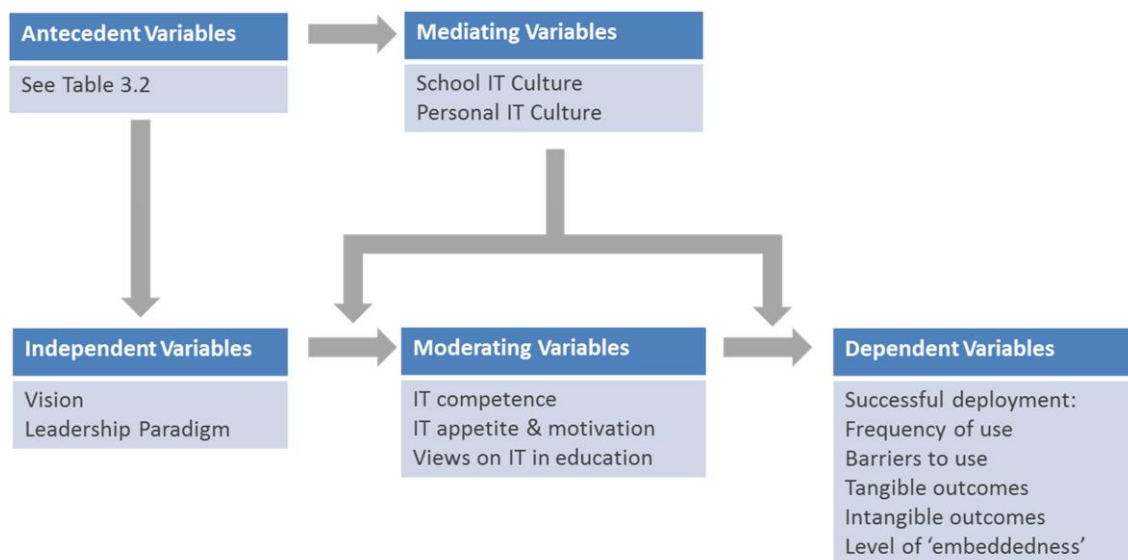


Figure 3.1 – Research Framework, based on Leithwood & Levin (2005, p. 13)

3.3 Research Variables

The potential list of variables which might have contributed to the research is very large, and a significant effort has been applied in attempting to identify a comprehensive list of potential variables, from which key variables of interest would be selected.

Table 3.2 summarises this work, with variables not specifically included in the research assuming the role of antecedent variables, presented here to provide a sense of context, perspective and scope for the variables of interest. It is recognised that in another research project, or even with a minor adjustment to the focus of the current project, any of these variables might be re-categorised into a different role.

Factor	
Antecedent Variables	School goals: explicit, implicit, and the extent to which these are communicated and perceived to be clear, meaningful, useful, current and consistent. Organisational context: stakeholders, policy environment. Organisational attributes & culture: size, rewards, job enrichment opportunities, interpersonal dynamics, goal or task difficulty. Leaders: Gender, hierarchy, background, motivations, values School: Structure (eg size, age), philosophy & ethos, organisational learning processes, improvement processes, relationships with families, instructional policies & practices. Teacher: pedagogical competence & subject matter knowledge, classroom experience, general motivation. Class: size, teaching loads, homework, student grouping, curriculum & instruction. Pupil: family backgrounds incl. educational culture, gender, ethnicity, beliefs & values, morale, trust.
Independent Variables	Vision: Local/national Leadership style: Transformational, Managerial, Instructional, Participative
Dependent Variables	Successful Deployment of Technology: frequency of use, barriers to use, tangible outcomes, intangible outcomes.
Moderating Variables	Deployment & Integration
Other Moderating & Mediating Variables	IT attributes: Competence, IT appetite & motivation, Views on the role of technology in education, Access to e-content, School Culture: IT impact, Personal culture: IT impact,

Table 3.2 – Principal Variables and their relationship to the Research Framework

Determining a precise role for each variable is by no means straightforward; some variables might equally act as moderator or mediator, depending on the context. The effect of variables upon each other appears interdependent and interrelated, with variables sometimes exchanging roles.

Roe (What is wrong with mediators and moderators?, 2012) examines precisely this dilemma and offers helpful guidance. He invites us to consider the temporal nature of variables, contending that in cross-sectional research a conventional moderator-mediator view of variable relationships may be misleading. He argues that variables can be multi-modal, with complex interrelated and reciprocal effects. The implications for research design modelling are significant; that variables do not necessarily retain fixed roles – they can be unstable and often unpredictable in nature.

For the author this is a more satisfying and pragmatic perspective, providing a degree of freedom from the fixed role of variables set out in Table 3.2, allowing more nuanced relationships and contextual freedom to be applied in the categorisation of research variables. The core notions of Independent, Dependant, Moderator and Mediator variables is retained, as is the research framework itself, but within a more relaxed and flexible philosophical model.

3.4 Research Strategy and Instruments

The primary instrument for this project consists of a survey of practicing educationalists in UK schools.

3.4.1 Research Population and Sampling

It was not possible to obtain a database providing personalised access to every single teacher, and even if that had been the case, it would not have been feasible or desirable to engage in such a mammoth project. Instead, a publicly available database providing email addresses of over 22,000 UK schools was obtained, with the objective achieving a response rate of perhaps 2-3% from this population.

This original email list was in Microsoft Excel format and contained a wide range of additional data pertaining to the schools. It was noted that a substantial number of the addresses were of the form ‘head@’, ‘office@’, ‘admin@’ etc., so it was anticipated that a high proportion might be processed by admin (or non-teaching) staff, thus increasing the potential for non-response.

From the list it might have been possible to devise a sampling frame to pre-select only named addressees, or choose schools according to type or location, and so on. However, this form of pre-selection would be affected by potential uncertainties arising from any inaccuracies within the database, along with the practical realities of exactly who in the school would be opening the '*request to participate*' emails. So no pre-selection has been applied and hence, the nature of this research could be regarded as a census of UK schools, rather than a survey of a carefully selected sample from the population (Saunders, Lewis, & Thornhill, 2012, p. 261).

The list was manually edited to extract the email addresses and discard other data, partly to help preserve school and participant anonymity by isolating the email address from school name and other contact information, and also to fragment the single file into several sub files.

These sub-files allowed pilot testing of the questionnaire to five sub-groups each consisting of 500 email addresses, followed by five 'live' questionnaire distributions, each to approximately 4000 email addresses.

The questionnaire was designed to take less than 10 minutes for a typical respondent to complete, based on research (Saunders, Lewis, & Thornhill, 2012, p. 453) that questionnaire length has an inverse relationship to response rates.

Research ethics, according to British Psychological Society guidelines were incorporated (BPS, 2014), as set out in Annex 4.

3.4.2 Questionnaire Pilots

The questionnaire was piloted to five sub-groups of email addresses, selected by re-sorting the email list into alphabetical order, assigning a repeating, sequential ID in the range 1 to 50, then extracting email addresses according to this ID into a sub-file. The large number of email addresses in the list afforded the author the luxury of being able to undertake multiple pilot surveys, each in itself of a reasonable scale, to thoroughly fine tune the technical aspects, cover letters, question set, and the format and analysis of the resulting data. A summary of the pilot objectives is set out in Table 3.3.

	Objective
1	Test and evaluate the process of survey creation, distribution, response monitoring and data analysis.
2	Determine possible response levels, to judge the correct balance between response rate and level of detail.
3	Understand the types of respondent (ie role in the school), and how this might impact on survey design and results analysis.

Table 3.3 – Objectives of the Questionnaire Pilots

Two online ‘survey engines’ were evaluated during the pilots; SurveyMonkey and Toluna QuickSurveys, with widely differing response rates. A summary of the key learning points arising from the pilot surveys is presented in Table 3.4, with Annex 3 providing additional information on the piloting and survey process, in terms of the number of questionnaires distributed at each stage, the response rates, and the learning points to feed into the next stage.

	Learning Points
1	The response level from SurveyMonkey was significantly higher than QuickSurveys. The data was very consistent across all five pilots, suggesting that more of the emails delivered by SurveyMonkey actually made it into respondents Inbox’s as compared to those from QuickSurveys.
2	The cover letter needed slight amendment to ensure that respondents had a clear indication of the purpose of the research (and hence, were motivated to participate), though some descriptive wording was omitted as it was felt that this might be influencing respondent choices.
3	A large proportion of respondents were head teachers, not classroom teachers. The cover letter was amended to strongly request that the research invitation was circulated to colleagues throughout the school. (This subsequently yielded a 30% increase in responses from practicing classroom teachers).
4	Offering respondents the possibility of receiving a summary of the results did seem to increase participation levels, and the offer was included in the ‘live’ surveys.
5	Sending out the initial invitation midweek (avoiding Mondays and Fridays) seemed to yield a higher response rate.
6	Sending out a follow-up reminder email one week later yielded an additional 40-50% response rate.
7	The number of questions in the survey was reduced, resulting in a 30% increase in respondents who completed the whole questionnaire, rather than quitting part way through.

Table 3.4 – Learning Outcomes from the Questionnaire Pilots

The pilots proved to be an essential aspect of the overall process, not simply in ensuring that the questionnaire actually reached as many of the target group as possible, but in improving the communication to participants so they were motivated, yet not influenced in their responses, either by the cover letter or the structure of the questionnaire itself.

3.4.3 Questionnaire Structure

The overall design philosophy for the questionnaire was to commence with a series of easy questions which were unlikely to present a barrier for any respondent, with the more difficult questions requiring greater consideration coming closer to the end. In this way, many respondents would have made a time and effort commitment to the process when they reached the trickier questions, reducing the temptation for them to bail out.

Questions were divided into sub-sections, so that respondents could get a sense of context and flow during their progression. Table 3.5 summarises the sections forming the questionnaire, and a fully comprehensive Data Requirements Table (Saunders, Lewis, & Thornhill, 2012, pp. 425-8) is presented in Annex 6.

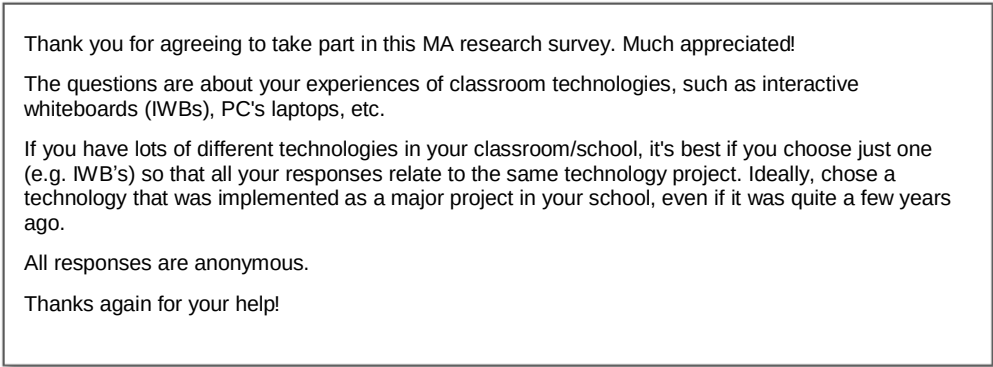
Section	Topic	Number of Questions
1	The respondent	3
2	The classroom technology project being considered	7
3	Strategy & vision in relation to the project	5
4	Project management & implementation	5
5	Change management	4
6	Vision and strategy	13
7	Technology impact & outcomes	8
8	School culture	8
9	Leadership	24
10	Survey feedback questions	3

Table 3.5 – Questionnaire Sections

The majority of research questions were structured as Likert scale responses from one to five; representing Strongly Disagree to Strongly Agree, with additional variables (such as role, experience, etc) being captured for analysis purposes.

One of the advantages of a tool such as Survey Monkey is that it automatically formats the questionnaire responses into data files which can easily downloaded and imported into spreadsheets, for offline sorting, coding and analysis.

Respondents were requested to select a specific technology project as the basis for their responses, to ensure that, so far as possible, all respondents were envisaging a large scale project affecting the whole school, rather than the introduction of a more specific item of technology which might have been a personal initiative or small scale trial. This was achieved through a simple instructional introduction on Page 1 of the Survey (Figure 3.2).

A rectangular box with a thin black border and a light gray drop shadow, containing the following text:

Thank you for agreeing to take part in this MA research survey. Much appreciated!

The questions are about your experiences of classroom technologies, such as interactive whiteboards (IWBs), PC's laptops, etc.

If you have lots of different technologies in your classroom/school, it's best if you choose just one (e.g. IWB's) so that all your responses relate to the same technology project. Ideally, chose a technology that was implemented as a major project in your school, even if it was quite a few years ago.

All responses are anonymous.

Thanks again for your help!

Figure 3.2 is a screenshot of the instructions provided to survey participants. The instructions are contained within a rectangular box with a thin black border and a light gray drop shadow. The text inside the box is as follows: 'Thank you for agreeing to take part in this MA research survey. Much appreciated!', 'The questions are about your experiences of classroom technologies, such as interactive whiteboards (IWBs), PC's laptops, etc.', 'If you have lots of different technologies in your classroom/school, it's best if you choose just one (e.g. IWB's) so that all your responses relate to the same technology project. Ideally, chose a technology that was implemented as a major project in your school, even if it was quite a few years ago.', 'All responses are anonymous.', and 'Thanks again for your help!'.

Figure 3.2 – Instructions to survey participants

4 Research Analysis

4.1 Questionnaire Responses: Overview

With 1603 responses to the online survey, the level of participation is felt to be very satisfactory given that:

The questionnaire was delivered to respondents as an unsolicited email;
In the majority of cases, the email address was a 'generic' office or admin function not a named person;
The email address list is publicly available, and hence, is likely to be widely used; these inboxes potentially receive lots of unsolicited junk mails, and this request could easily have been regarded as such and ignored/deleted.

Despite these constraints, the survey generated an overall response rate of 7.8%, which compares very favourably to reports of similar surveys conducted by others in the education sector, where response rates of 2-3% to unsolicited, broadcast surveys are considered good.

Not all responses were found to include answers to the entire question set, and it was clear that some respondents had decided to abort the process before reaching the end. Whilst the British Psychological Society guidelines (2014) suggest that only completion levels of less than 50% need be deemed as withdrawal from participation, for this research, only fully completed questionnaires were incorporated in the data analysis.

Hence, from the total of 1603 responses received, 544 were discarded, even though many of these were completed to a level in excess of 80%. According to Saunders et al (2012) sample sizes of 1056 and 1066 are required for statistical inference level of 3% margin of error for populations of 100,000 and 1,000,000 respectively, suggesting that the remaining sample of 1059 responses would be statistically representative of a large (potentially national) population of teachers.

This of course is from an entirely theoretical perspective, and assumes a random and representative range of participants. This cannot be guaranteed given the format of the email address list, so elements were incorporated into the questionnaire design to help understand sub-groups that individual respondents belonged to, and whether they were close enough to the topics of the research to offer meaningful and valid responses.

This contextual information (Table 4.1) is highly valuable in understanding the background to individual responses, and during the analysis process, provides the potential to introduce filtering mechanisms for more detailed exploration of the dataset.

	Analysis
Role	69% of respondents are head- teachers, with the remaining 31% being teachers or teaching assistants.
School Category	70% of respondents are in the Primary School sector, 22% in Secondary, and 8% in a mix of institutions such as Special Needs Schools or Colleges.
Years of Teaching Practice	The average number of years of teaching service in the sample is 20, which reflects the high number of head teachers participating in the research.
Questionnaire Length	70% of respondents felt that the Questionnaire length was just right, 24% felt that it was too long, and 6% would have liked more questions!
Questionnaire Relevance	74% of respondents felt that the questions were relevant to their educational context, with 8% indicating that the questions were highly relevant. Only 17% felt that the questions were not specifically relevant.

Table 4.1 – Basic information regarding the respondent context

These criteria allow responses from school Admin staff to be filtered out, and there is the potential to analyse data according to whether the respondent is a class teacher or head teacher, number of years of teaching experience and according school type.

Three questions in the survey enabled participants to provide feedback on the questionnaire itself, which facilitated fine tuning during the piloting process, and also helped to judge the integrity of the dataset.

For example, had a high proportion of respondents indicated that the questionnaire was overly long (leading to a loss of interest or proper consideration of the question/response), or not was really relevant to their educational context, then the pertinence of the dataset and its potential for analysis might have been questionable.

Thankfully, less than 25% of respondents felt that the questionnaire was too lengthy (and even these achieved 100% completion), with 81% indicating that the content was relevant or highly relevant to their educational context. Furthermore, 12% of respondents have requested a copy of the research summary, suggesting that the outcomes of the project are of significant interest to many in the education community.

Hence, we can reasonably conclude that the database is competent and provides a viable basis for analysis.

4.2 Questionnaire Analysis Process

Having downloaded the data (from the Survey Monkey website) and merged the separate response files into a single Excel database, the first phase was to ensure data integrity by summing the contents of a random selection of 10 data columns from each sub-file and cross-checking these against the consolidated values in the composite database.

Once successfully merged and cross-checked, the resulting validated database consisted of 1059 data rows each with 85 data columns; a total of 90,000 discrete data points.

The second phase of the analysis process required that data relating *directly* to the key research variables be isolated and subjected to a preliminary review. The purpose of the preliminary review is to gain an initial understanding of whether there are very clear and prominent indications arising from the data, or whether the data are more complex, perhaps with nuanced and subtle interrelationships requiring sophisticated forms of analysis.

The main dependant variables being studied in this research are those which relate to the outcomes of classroom technology projects. So in order to gain an outline understanding of the spread of results relating to these variables, the relevant Likert data points in the composite database were grouped and ordered in an Excel graph.

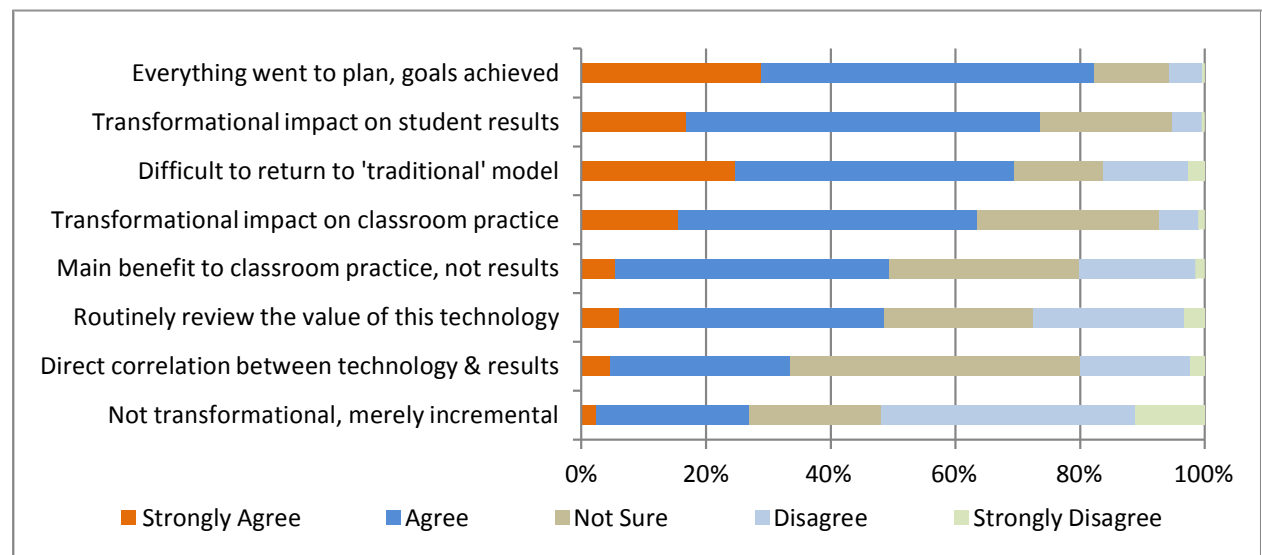


Figure 4.1 – Technology Impact Analysis $n=1059$. [58,355/40]¹

¹ Graphs in this dissertation use data which is one of: *base* (original database), *intermediate* (aggregated *base* data) or *derived* (calculated using *base* or *intermediate* data). A count of the number of each type of data item used in the graph has been included with the figure descriptions to give a sense of the data represented, using the notation *nnn/nnn* for *base/derived* and *nnn/nnn/nnn* where *base/intermediate/derived* figures are used.

The resulting graph (Figure 4.1), with elements arranged in 'level of agreement' order provides some fascinating insights into the impact of technology in the classroom:

The majority of respondents indicate agreement across several measures that the technology project forming the subject of the questionnaire has achieved success;

- 82% agree that their technology project went according to plan, and has achieved its objectives,
- 74% agree that the technology has had a transformational effect on student results,
- 70% agree that it would be difficult to return to a 'traditional' teaching model, where the technology was not available,
- 64% agree that the technology has had a transformational impact on their classroom practice, and their professional development as teachers.

The response to an 'alternative form' check question (Mitchel, 1996) to assess whether respondents felt that the technology had merely had an incremental impact, not transformational, yielded dis-agreement of 27%, suggesting that the positive responses to the transformational questions are valid.

Only 34% of respondents believe that the impact of technology project is measurable in terms of improved student results, and that there is a direct correlation;

A surprising 49% regularly review the ongoing value of the technology.

From this preliminary review of the dependant research variables, it is clear that there are factors of genuine interest embodied within the research data, and that systematic assessment of other variables in relation to these outputs is viable.

5 Research Findings

This section presents a detailed exploration the data following a structured approach. Each subsection addresses a specific topic, considering data elements which together comprise the dependent, independent, moderating and mediating variables which will form the final, overall analysis. In each subsection there is a graphical analysis of important findings to provide context and easily assimilated comparisons, followed by a statistical summary in a format which feeds directly into the final research analysis model (Section 5.7), according to the framework described previously in Figure 3.1.

5.1 Overview of Leadership Opinions

The leadership models of interest arising from the literature review are Transformational, Instructional, Managerial and Participative. Multiple aspects of the questionnaire sought to reveal respondents views of the prevailing leadership styles in their schools, with additional information relating to school culture also being investigated, given that these might represent *proxy* indicators of leadership, from the effects of the leadership style on school culture.

The questionnaire data was grouped according to leadership typology, as set out in Annex 7, which seeks to categorise leadership behaviours in terms of their direct or inverse relationship to each paradigm. Annex 7 provides a good illustration of the complexity in analysing the underlying data to form specific deductions, due to the high number of data points and the troublesome nature of leadership definitions; that several characteristics apply in more than one definition.

It is an approach which bears some similarities to the work of Currie and Lockett (2007) who coded key words found in the narrative of OfSTED reports in an attempt to extrapolate the leadership paradigm.

However, it is arguable that the approach summarised in Annex 7 is significantly more robust than Currie and Lockett's method, as the range of questions is broader, exactly the same terminology is used with all participants, and data for exactly the same question set is available from all respondents. Furthermore, the Annex 7 approach includes both direct and inverse characteristics for each paradigm.

In the initial phase of analysis the Likert responses to questions of ‘*who is responsible for specific aspects of school leadership*’ were isolated and arranged for graphical presentation (Figure 5.1). In this form, opinions of the distribution of leadership activities from head through to individual staff members can be clearly observed.

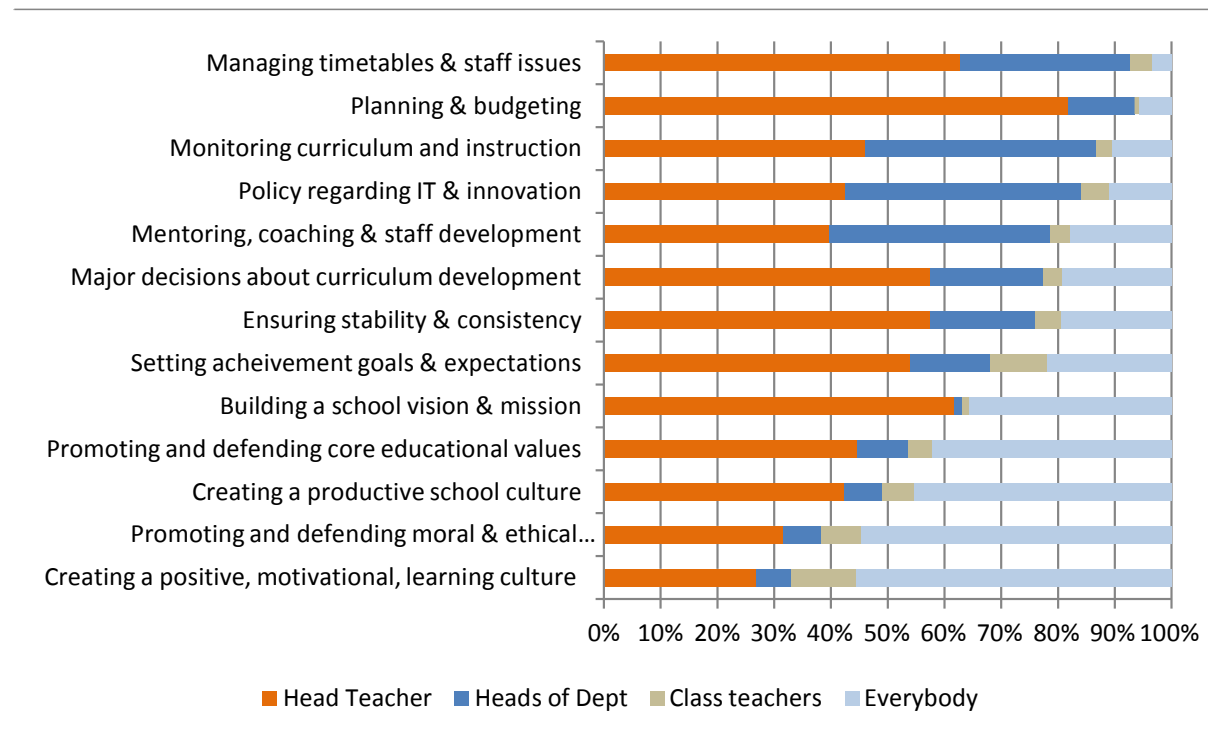


Figure 5.1 – Distribution of Leadership function $n=1059$ [13,767/52]

This analysis, superficially at least, suggests that Managerial styles of leadership are somewhat dominant (supporting Currie and Lockett, 2007), given the high incidence of administrative functions which are considered to be the responsibility of the head, or heads of department.

There would appear to be a clear tendency for aspects of vision, ethos, culture and learning motivation to be ‘pushed down’ to staff as a shared responsibility, rather than being considered a specific role of the head.

However, a parallel analysis isolating those functions which are *explicitly* considered to be the responsibility of the head (Figure 5.2) reveals some interesting contradictions with the data in Figure 5.1, questioning of the accuracy of Currie and Lockett’s findings.

These effects may be due to differences in the question design, as the Figure 5.2 questions are specifically about the responsibilities of the head. So in a sense it is more a measure of the perceived degree of *responsibility* attached to each function; the more *responsible* a function is considered, the more likely a respondent may be to assume it to be the purview of the head.

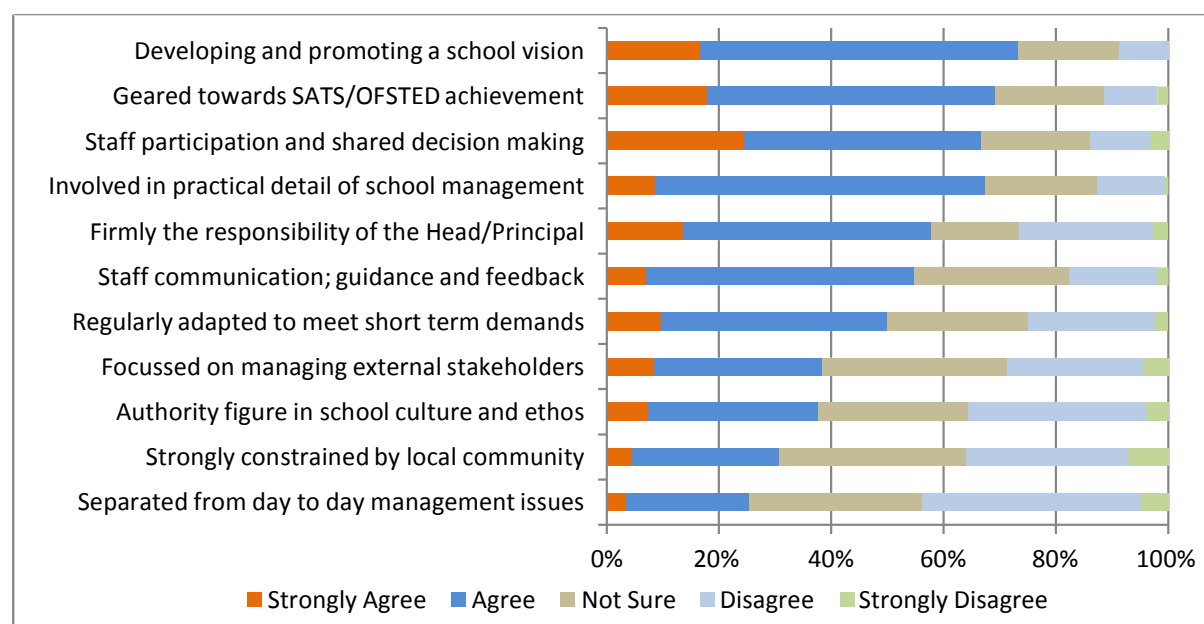


Figure 5.2 – Opinions of Leadership responsibility $n=1059$ [58,245/55]

A third data series within the dataset also contains views on leadership, but more precisely, they are views of school culture, and hence, might be considered as a *proxy* measure of leadership through the impact that the leadership style has on the school culture.

Figure 5.3 summarises this data, which suggests that Participative styles of leadership may have a significant effect on school culture, particularly when compared to those measures which may be indicative of a more *managerially* oriented leadership paradigm.

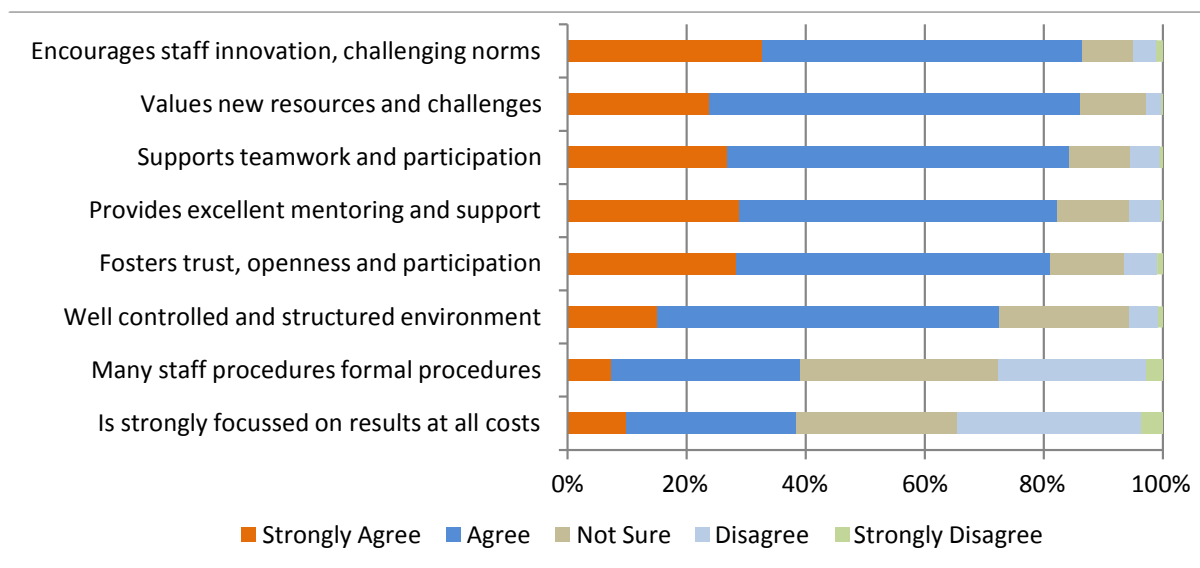


Figure 5.3 – School Culture as a proxy for Leadership style $n=1059$ [8,472/40]

5.2 Isolating Leadership Paradigms

Determining the dominant paradigm applicable to each respondent's school context is fundamental to this research. There are several stages, beginning with the processing and filtering of the raw response data using the methodology described in Table 5.1, which requires a matrix transform to be constructed to adapt the 1 to 5 numerical Likert response.

The inclusion of positive and negative values in the matrix transform results in a positive to negative range in the resulting 'scores', with the negative scores reflecting those responses indicating *disagreement* to a given leadership characteristic, or where there is a view that some leadership tasks should be delegated downwards to staff, and are not the responsibility of the head.

	Data manipulation stages
1	For the responses to the 'School Culture' Questions, assign values of 2,1,0,-2,1 (Strongly Agree to Strongly Disagree)
2	For the responses to the 'Leadership Style' Questions, assign values of 2,1,0,-2,1 (Strongly Agree to Strongly Disagree)
3	For the responses to the 'Leadership Style' Questions, assign values of 2,1,0,-2,1 (Strongly Agree to Strongly Disagree)
4	For the responses to the 'Leadership Responsibility' Questions, assign values of 0,2,1,0,-1,0,-2,0 (Governors, Head, etc.... Everybody, Nobody)
5	Create a 'transform matrix' using these three sets of values, which is then applied to the leadership opinion data of all respondents;
6	Compute a composite leadership score for each paradigm by using the Leadership Typology data (Annex 7) to generate an analysis for each respondent of opinions relating to each of the leadership paradigms; Transformational, Instructional, Managerial and Participative.

Table 5.1 - Response data processing to adapt raw response data prior to statistical analysis

The aggregation of this data (Figure 5.4) illustrates the spread of opinions, ranging from disagreement (-11) to agreement (+11), indicating the extent to which respondents believe that particular characteristics are reflected in the leadership paradigms in their schools.

The area under each curve is the same, as each dataset contains data for the same total number of respondents.

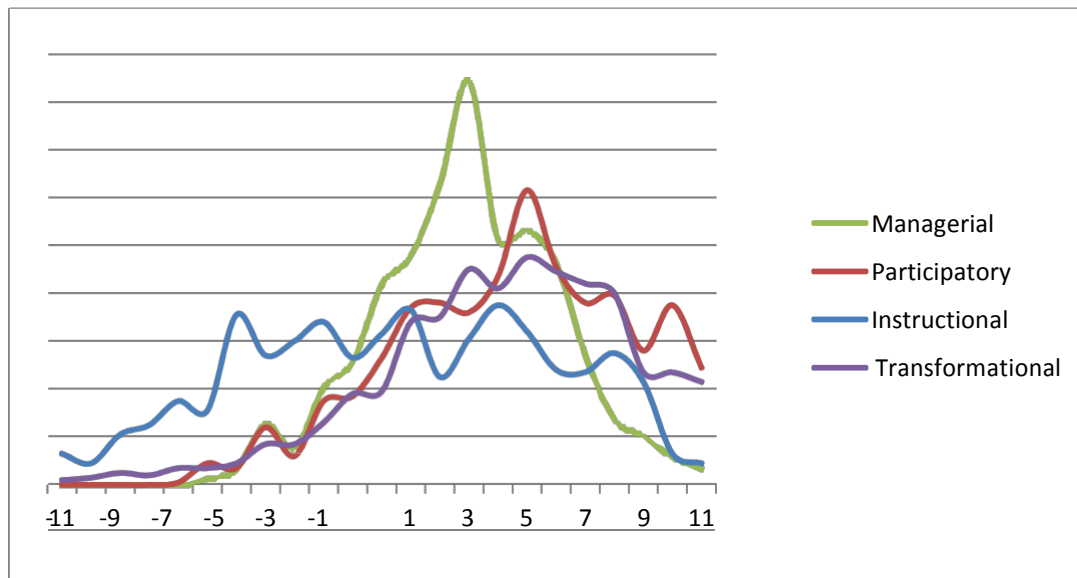


Figure 5.4 – Leadership Paradigm Opinions $n=1059$ [55,172/8,488/92]

A top level analysis of Figure 5.4 suggests:

The Managerial paradigm is the most consistently distributed style, with the majority of opinions grouped in the range -3 to +8;

The Participative paradigm has a broader distribution of opinion, with the number of positive opinions extending to +11, and the peak of positive opinion being at +5 as compared to only +3 for the Managerial style;

The Transformational paradigm has a similar broad distribution of positive opinion to Participative, though with a small indication of negative opinion all the way down to -11, the lowest rating;

The Instructional paradigm has the most divided set of opinions, with respondents views right across the spectrum from -11 to +11.

Applying a Kolmogorov-Smirnov test to these data (Table 5.2) confirms that the Instructional paradigm does not follow a normal distribution. Whilst further research would be needed in order to understand this more fully, a possible hypothesis might be that, when given a straight choice, more respondents prioritise the managerial and collaborative aspects of leadership over instructional aspects.

Paradigm	D n, α		
	Mean	SD	D max
Instructional	0.12	5.26	0.0491474
Participative	3.91	5.27	0.0384939
Managerial	2.38	3.15	0.0239403
Transformational	3.56	4.26	0.0415207

Table 5.2 – Kolmogorov-Smirnov analysis of Leadership Paradigm opinions $n=1059$

A summary of the overall opinion of all respondents to determine the precedence of leadership paradigms is revealed in Figure 5.5, where it is apparent that the dominant style is Transformational, closely followed by Participative, with Managerial and Instructional being clearly positioned in third and fourth place.

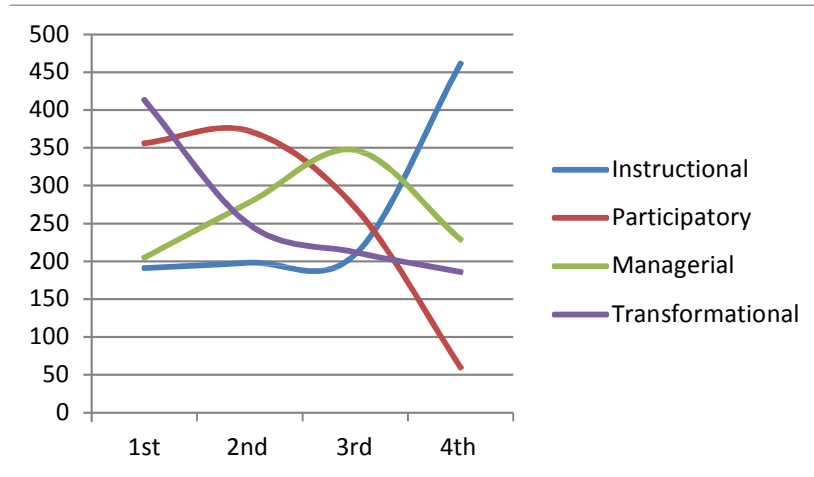


Figure 5.5 – Leadership Paradigm Ranking $n=1059$ [55,172/8,488/16]

From the Kolmogorov-Smirnov analyses we can be confident that the distribution of different groupings in the dataset is generally consistent (albeit with some notable exceptions), and that the dataset is largely homogeneous with respect to overall leadership paradigm opinion.

Having computed a ‘dominant leadership paradigm’ for each respondent, we can now investigate ‘upstream’ linkages between leadership style and vision, and ‘downstream’ linkages between leadership style and outcomes.

5.3 Project Vision

A recurring theme in the reviewed literature is the importance of vision in any leadership or change management context. Several of the questionnaire elements directly explore the issue of vision, whilst other questions investigate topics which infer or might be regarded as proxies for vision, for example, the motivations behind the technology projects.

The analysis objective in this section is two calculate, for each respondent, two independent variables which aggregate opinions relating to vision (Table 5.3).

	Independent Variables linked to Vision
Vt	Indicates the overall strength of the technology vision and its source.
Vr	Indicates the extent to which vision is regarded as being a predominantly leadership responsibility rather than being an assumed element of everyone’s role.

Table 5.3 – Independent Variables with respect to Vision

One key question is where the technology vision originated; whether this is developed internally by the school and its leadership, or instigated by external influences. The two extremes to be understood first of all are whether the technology vision has originated at national policy level or at school level.

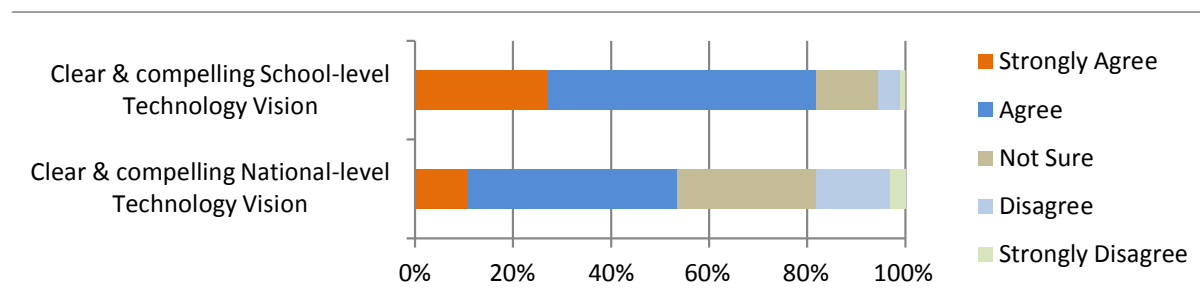


Figure 5.6 – National vs Local Technology Vision $n=1059$ [2,118/10]

The differences in perceptions are interesting and superficially suggest a more introspective approach to leadership and decision making for technology projects. However, it is common sense that, in reality, few schools make their technology decisions in complete isolation; most projects must be influenced, to some degree, by initiatives and developments in other schools.

Despite the logic of this thesis of external influence, only 53% of respondents (Figure 5.6) feel that their project connects to a national policy or strategy context, whereas a very substantial 82% of respondents state that the main drivers originate from within their own school.

This has important implications for school leadership and culture, and is a very significant factor in respect of the *ownership* of the vision. Ownership is central in avoiding potentially negative consequences arising from perceptions that the project may have been imposed on the school from external sources.

Yet, notwithstanding respondent's insistence that projects were mainly driven by internal school motivations rather than external influences, 57% of respondents indicated that the main project catalyst was indirectly connected to the success of similar initiatives elsewhere, because the technology benefits had been well established. (Figure 5.7).

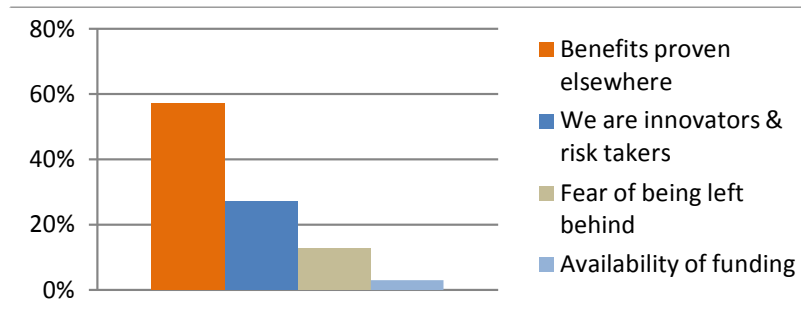


Figure 5.7 – Project motivations in relation to Vision $n=1059$ [1,059/4]

In addition, 13% of respondents cited fear of being left behind [by other schools] as a project motivation, which suggests that in aggregate, external factors may influence the primary motivations in around 70% of projects.

Only 27% stated that the intrinsic desire to for their school to innovate, take some risks and be educationally creative was the main driver.

So whilst asserting self-determination in respect of technology projects, there appears to be a more substantial element of external influence than many respondents would superficially indicate.

An important aspect of vision is the degree of understanding of the purpose and expected outcomes. In total, 82% of respondents indicated that prior to project roll-out, there was a wide consensus of expectation across the school staff that the technology would bring educational benefits.

This suggests that there are functional mechanisms to articulate a technology vision within these schools, and whatever the paradigm, this aspect of leadership is effective.

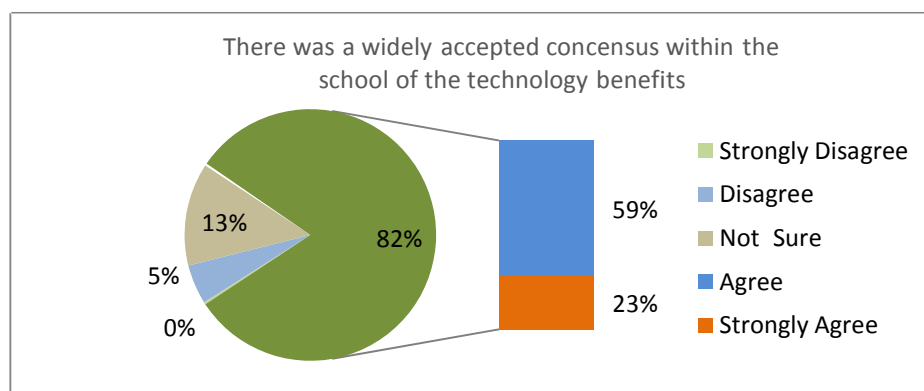


Figure 5.8 – Project motivations in relation to Vision $n=1059$ [1,059/5]

Respondent attitudes to future classroom technology projects might be regarded as a proxy for educational vision; the greater the appetite for technology, the more likely it is for schools to have a coherent, technology related school vision. This is explored in Figure 5.9, with 75% of respondents indicating a desire for more technology in the classroom. It is an important finding, suggesting that many schools see increased utilisation of technology as a core part of their vision, or rather, that improved educational outcomes might be realised through an ongoing technology strategy.

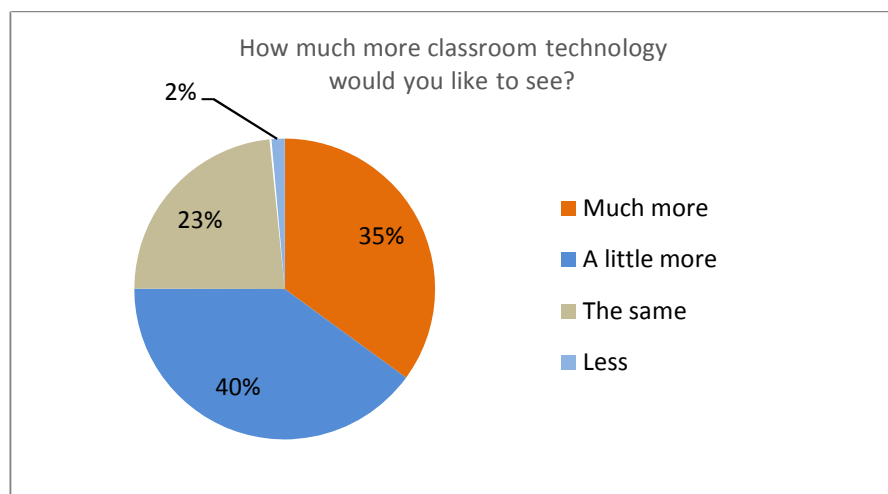


Figure 5.9 – Appetite for more classroom technology $n=1059$ [1,059/4]

Of perhaps greater significance is the question of where the primary responsibility for strategy and vision sits in the school, and hence, how these influencing factors might be connected with the schools leadership paradigm.

Figure 5.10 presents responses to a sub-set of the questions relating to school leadership responsibilities, and how these are distributed. The graph seeks to juxtapose two distinct components of school management; long term, related to strategy and vision, and short term, related to management of day to day issues.

Of particular interest we find that:

61% of respondents believe that the head teacher has the main responsibility for planning and budgeting, as compared to 55% who feel that the head teacher is mainly responsible for vision and mission; 32% suggest that developing the schools vision and mission is everyone's responsibility, rather than being primarily the role of the school leadership in the form of school governors and head teacher;

43% of respondents feel that IT strategy (as it relates to the schools vision) is the responsibility of governors and head teacher, whilst 37% indicate that this is a middle management function, to be left with heads of department.

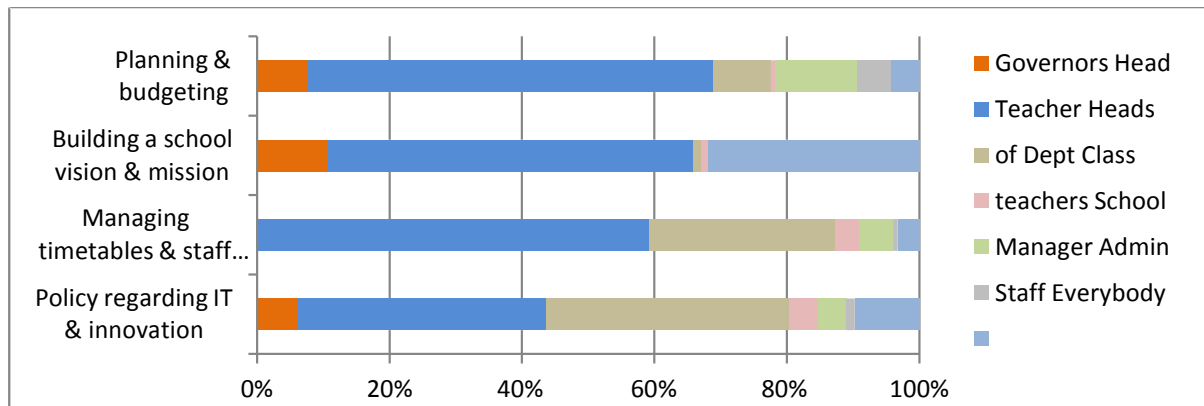


Figure 5.10 – Leadership roles in respect of strategy and vision $n=1059$ [4,236/28]

Respondents seem to indicate that vision and mission are not specifically a leadership function, certainly no more so than routine management activities.

Given that 60% of respondents are head teachers, it is interesting that the overall majority wish to share (or delegate) this aspect of school management amongst their junior colleagues. Similarly, less than half of respondents feel that IT strategy is sufficiently critical for it to be in the sole purview of governors and heads; it is more of a tactical issue, to be delegated to heads of department, individual teachers and admin staff.

Variable	Opinion Factor	Mean	SD
Vt	Appetite for additional technology	4.07	0.82
	Compelling national IT vision	3.41	0.98
	Compelling school IT vision	4.00	0.85
	Consensus on IT benefits	3.91	0.94
	Compelling school IT vision	4.02	0.88
	Consensus on IT benefits	3.90	0.84
	Project Motivation	3.71	0.89
Vr	Vision is the responsibility of the head	2.79	1.96
	IT & innovation are leadership responsibility	3.03	1.36
	The heads role is primarily focused on vision	3.81	0.82

Table 5.4 – Analysis of Vision opinion responses $n=1059$

Table 5.4 summarises the findings from this section in terms of the independent variables which aggregate the two key aspects of vision studied in this research.

The distribution of responses for V_t shows good consistencies, finding generally positive agreement regarding the strength of the technology vision in schools, with a mean of 4 indicating ‘Agree’ on the Likert response. The exception being the views on nationally inspired motivations, discussed previously in relation to Figure 5.6.

The distribution of responses for V_r is less homogeneous, reflecting both the ambiguity of opinion regarding leadership responsibility for vision (3 = “neither agree/disagree”), combined with those views that vision and mission are “everyone’s responsibility”.

A Cronbach’s Alpha analysis indicates weak internal consistency for data relating to vision responsibility, suggesting that the variables comprising V_r are measurements of a range of dissimilar factors.

	Project Vision – Cronbach’s Alpha
V_t	0.6857
V_r	0.1894

Table 5.5 – Project Vision – Cronbach’s Alpha

5.4 Technology Integration Process

Much of the literature supports the view that it is the effectiveness of integration which largely determines the extent of successful outcomes in educational IT projects.

The objective in this section is to calculate, for each respondent, three moderating variables which aggregate opinions of the technology integration process (Table 5.6). They are measures of how well the project design sought to ensure effective integration, as opposed to post-project measures of actual integration, which are considered later as *outcomes*.

	Dependent Variables linked to Technology Integration
Tlp	Aspects of integration which are associated with project planning
Tls	Aspects of integration which are associated with staff views
Tit	Aspects of integration which are associated with the project team

Table 5.6 – Moderating variables with respect to Integration

Several measures were incorporated into the questionnaire to explore integration, beginning with the degree of consultation at project inception, when the technology vision was initially articulated to staff.

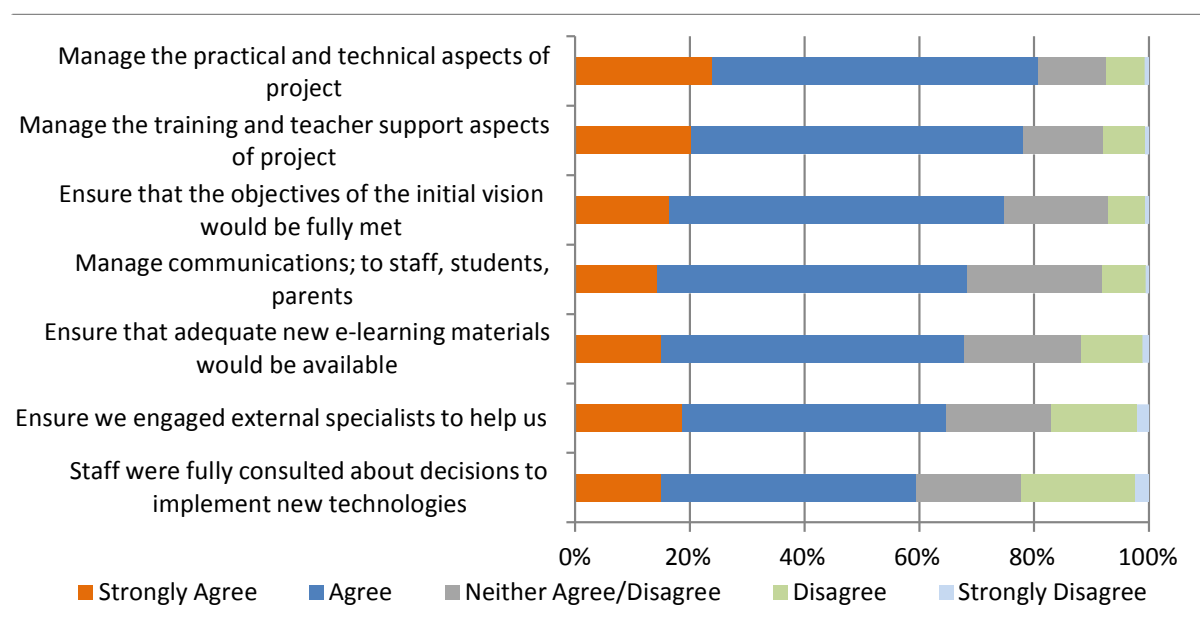


Figure 5.11 – Project planning issues in relation to Technology Integration $n=1059$ [7,413/35]

For the majority of project planning topics (Figure 5.11) we can see there are high levels of agreement, suggesting that it is commonplace for integration to be properly considered and accommodated within the project design. There is a slightly reduced consensus with respect to communication of project objectives and the provision of adequate e-learning materials. The use of external specialists may not have been universally necessary, and to a degree, this may not have been a fair question as respondents are considering this matter in hindsight, where some may wish that they had foreseen the need for additional expertise.

The most significant issue arising from the analysis in Figure 5.11 is that of staff consultation. Whilst 60% agreement to the notion that staff being fully consulted is commendable, that still leaves a very significant proportion where there is uncertainty or disagreement. As we know from the general literature on change management and the specific literature on education technology projects, consultation and ‘buy-in’ is an important factor in achieving the desired outcomes.

Staff reactions to the proposed changes which the technology project might bring are explored in Figure 5.12, and reveal several areas of interest. In particular, high levels of staff enthusiasm for the project at 73%, with only 22% of respondents suggesting that the project caused tensions and conflict at the planning stage.

Similarly, in the majority of cases there were no major concerns or the concerns were properly discussed and accommodated, with only 19% and 7% respectively not agreeing that this was the case.

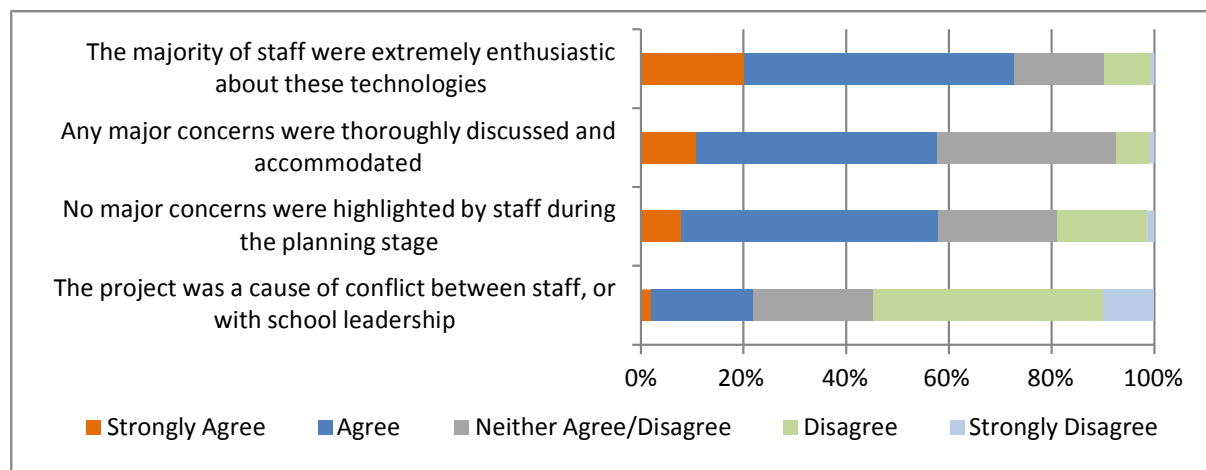


Figure 5.12 – Initial staff reactions in relation to Technology Integration $n=1059$ [4,326/20]

Figure 5.13 explores the question of whether the project was implemented as a cohesive team, sharing goals and expectations through effective alignment with the leadership, as opposed to a group of individuals each with their own personal agendas.

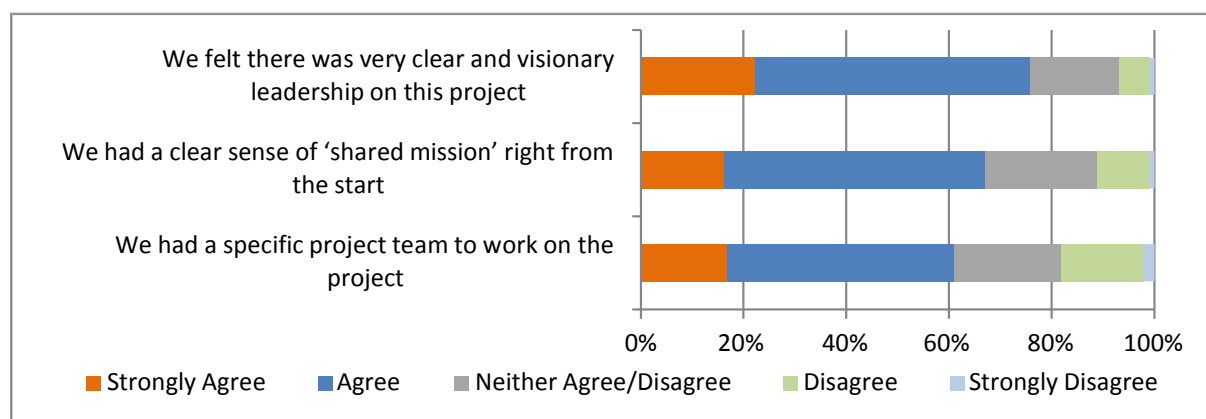


Figure 5.14 – Staff collaboration in relation to Technology Integration $n=1059$ [3,177/15]

From Figure 5.13 we can see that in the majority of cases there is strong agreement that the concepts of leadership, team working and shared mission were inherent elements of the projects. These findings further support the view that significant efforts were applied in the majority of cases to ensure effective technology integration.

Taken in aggregate, these data present a very positive picture of issues related to effective technology integration. There is a slight concern with regard to the issue of staff conflict during the planning process, though this is a normal and expected aspect of change.

Variable	Opinion Factor	Mean	SD
Tip	Staff Consultation	3.50	1.05
	Engaged external specialists to help us	3.64	1.01
	Adequate new e-learning materials	3.70	0.89
	Manage communications	3.74	0.82
	Ensure that the objectives would be fully met	3.83	0.80
	Training and teacher support aspects	3.90	0.83
	Training and teacher support aspects	3.90	0.83
	Practical and technical aspects of project	3.97	0.83
Tis	Project was a cause of conflict	2.59	0.58
	No major concerns highlighted	3.45	0.91
	Major concerns were accommodated	3.60	0.80
	Staff were extremely enthusiastic	3.82	0.88
Tlt	Dedicated project team	3.58	1.01
	Sense of 'shared mission' from the start	3.71	0.89
	Clear and visionary leadership	3.90	0.84

Table 5.7 – Analysis of Vision opinion responses $n=1059$

A Cronbach's Alpha analysis indicates strong internal consistency for the variables associated with planning and team inter-relationships, with weak consistency for views of the project in its early stages.

	Technology Integration – Cronbach's Alpha
Tip	0.8902
Tis	0.4998
Tlt	0.866

Table 5.8 – Technology Integration, Cronbach's Alpha

5.5 Outcomes: Dependent Variables

Measurement of outcomes – the *dependent variables* of this research - is somewhat problematic. Ideally, it would have been preferable to somehow generate dependent variables from quantitative measures of educational effectiveness, for example, from SATs results, exam grades or OfSTED performance tables. But this would have been an enormous undertaking, well beyond the scope of this research.

So to provide indications of technology project outcomes the questionnaire sought respondent opinion on three basic outcome categories; educational, teacher and school. These form the three dependent variables of the analysis (Table 5.9).

	Dependent Variables linked to Outcomes
Oe	Opinions of education related project benefits (outcomes)
Ot	Opinions of teacher related project benefits (outcomes)
Os	Opinions of school related project benefits (outcomes)

Table 5.9 – Dependent Variables – Project Outcomes

Figure 5.14 summarises the underlying data from which these variables are to be derived, revealing generally positive opinion of overall outcomes, but uncertainty as to specific, measurable outcomes in terms of student results.

Technology adoption would appear largely successful, with 69% agreeing that it would be difficult to return to a traditional classroom teaching model.

A similar number of respondents agreed that the project objectives had been achieved, with a consensus that classroom practice is one of the main benefits; from both a class and a teacher perspective in terms of professional development.

Whilst 73% of respondents agreed that the technology had realised a “transformational impact” on student learning, just 33% believed that it would be possible to correlate this “transformational impact” with quantitative, measurable, student achievement results. Only a small majority (51%) agree that the technology represents a transformational change, as opposed to being merely incremental.

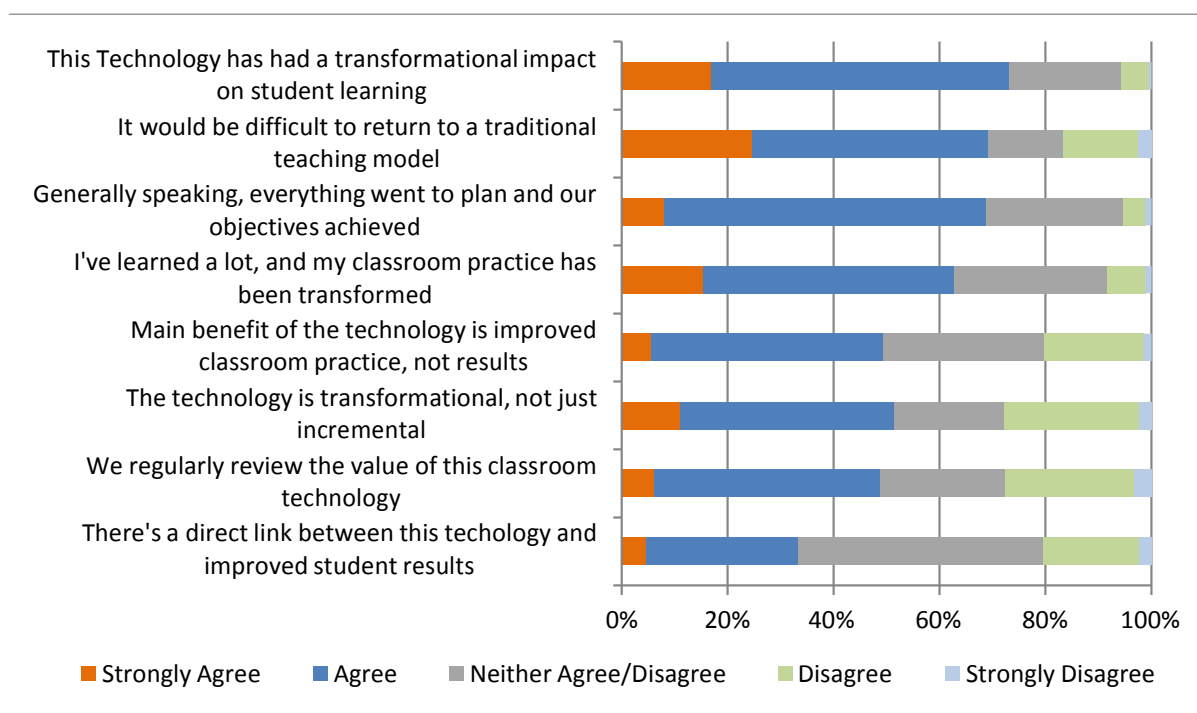


Figure 5.14 – Staff collaboration in relation to Technology Integration $n=1059$ [8,472/40]

Aggregating and categorising these views to derive the three dependant variables is straightforward, with the only point of note being that the question relating to ‘overall achievement of objectives’ is somewhat ambiguous; it has therefore been included in all three of the variables, as the projects may well have been designed with a central educational goal, whilst recognising the additional benefits to the school and to individual teachers.

Variable	Opinion Factor	Mean	SD
Oe	Transformational impact on student learning	3.84	0.78
	Everything went to plan, objectives achieved	3.70	0.72
	The technology is transformational	3.32	1.04
	There is a direct benefit to results	3.15	0.85
Os	Difficult to return to a traditional model	3.75	1.06
	Everything went to plan, objectives achieved	3.70	0.72
	Main benefit is improved classroom practice	3.33	0.89
	We regularly review the value of technology	3.24	0.99
Ot	I've learned a lot, my practice is improved	3.69	0.85
	Everything went to plan, objectives achieved	3.70	0.72

Table 5.10 – Analysis of Outcomes $n=1059$

The data shows that whilst there is variability within each outcome category, there is a clear indication that the benefits of IT projects to teachers are widely considered to be greater than they are to directly improving educational outcomes, as are the wider school benefits.

5.6 Moderating and Mediating Variables

The definition of moderating and mediating variables in this research framework is somewhat problematic, as previously described in the methodology section. The overall aim is to understand the influence of any variables which might impact the relationships between the independent variables and the dependant variables. Whilst the esoteric distinctions between moderating and mediating variables might be somewhat blurred in this context, the variables being explored fall into three categories (Table 5.11).

	Mediating and Moderating Variables
MV _{it}	Expressions of competence in information technology
MV _{ic}	Indications of teacher, or individual culture
MV _{sc}	Indications of school, or group culture

Table 5.11 – Mediating and moderating variables

There are indications in the literature that staff IT competence can be a significant factor in effective utilisation, both in terms of skills and in general attitude and confidence.

Respondents' views of IT and IT skills were triangulated from several dimensions; including competency, appetite for more classroom technology, degree of utilisation of existing IT, and the more philosophical question of whether IT is best used to support traditional teaching models or to experiment with radically new ones.

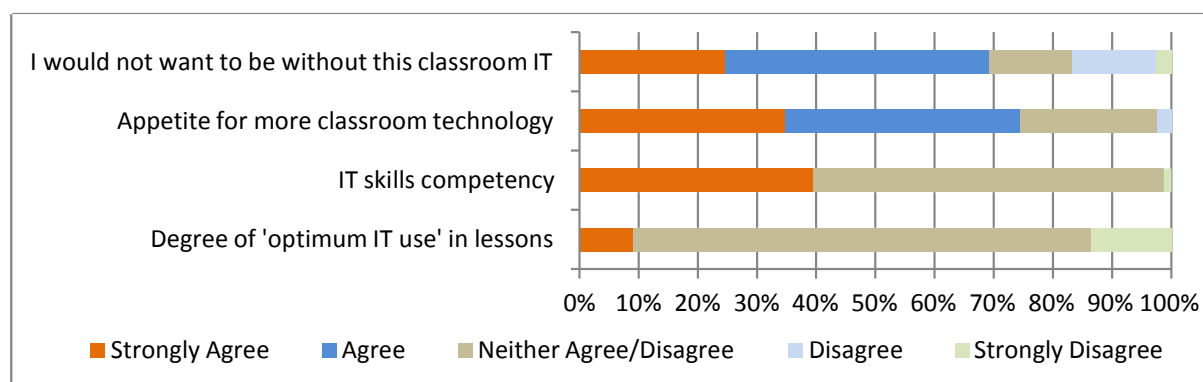


Figure 5.15 – Views of classroom technology. $n=1059$ [4,326/20]

The sample shows generally positive opinion across all factors (Figure 5.15), though interestingly, 69% of respondents feel that the role of IT should be restricted to supporting traditional pedagogical models.

It is a factor which links closely with more general teacher opinion which might be described as a *personal IT culture*, which is explored in Figure 5.16. Once again, there is a positive majority opinion, with only 5%, 7% and 15% respectively feeling that the technology has not had a transformational impact, from which their own professional development had not benefitted, or would want to return to a traditional, non-technology classroom model.

Only the question of concerns and risks regarding the technology deployment showed a slightly increased tendency towards the negative response, reflecting the natural concerns often expressed in change-related projects.

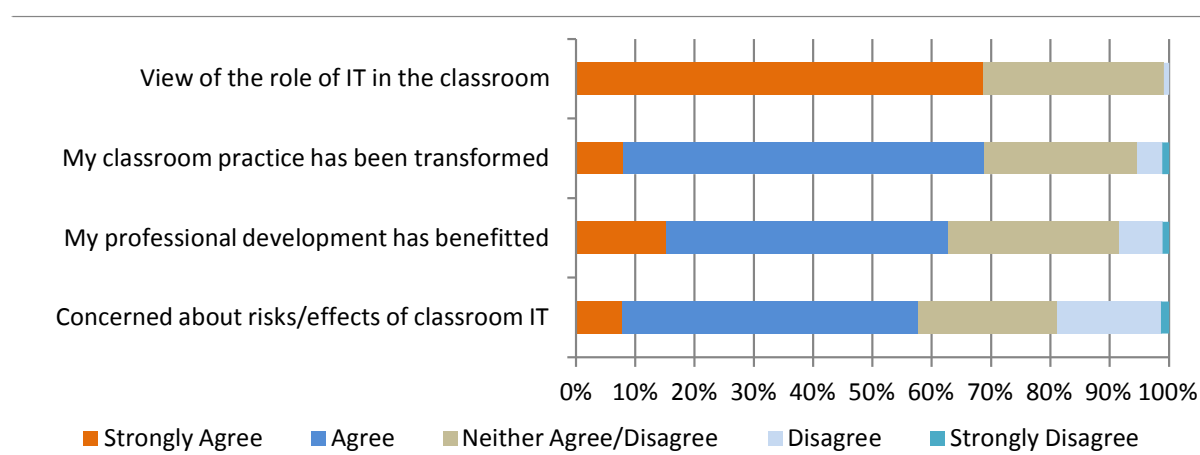


Figure 5.16 – Teacher culture regarding classroom technology. $n=1059$ [4,326/20]

Cultural influences at the school level were also explored, through questions relating to the overall level of staff enthusiasm for the IT project at the planning stage, whether the school as a whole shared a common sense of purpose, and whether the ongoing role of classroom technologies is regularly reviewed.

Figure 5.17 provides an overview of the responses; showing a good level of positive enthusiasm for the project at 70%, though with almost 40% unsure about there being a common purpose, and with less than half of respondents participating in a regular IT review.

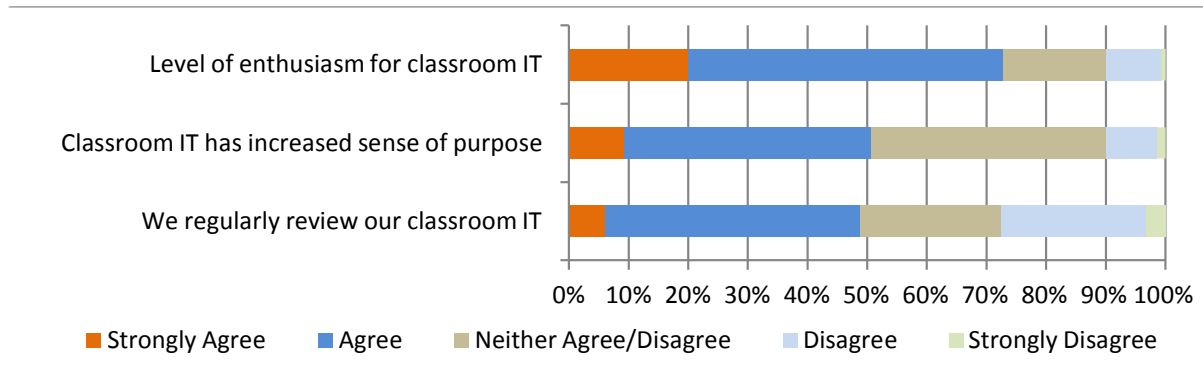


Figure 5.17 – School culture regarding classroom technology. $n=1059$ [3,177/15]

Together, these moderating and mediating variables *may* have an influence on how effectively the project leadership and vision translate into successful outcomes. Of particular interest is the variable expressing opinion on the general role of classroom IT; whether it is best used to support traditional teaching models, or as a catalyst and platform to drive the development of new paradigms.

Variable	Factor	Mean	SD
MV _{it}	Degree of 'optimum IT use' in lessons	2.91	0.94
	IT skills competency	3.77	1.02
	Appetite for more classroom technology	4.07	0.82
	I don't want to be without this classroom IT	3.75	1.06
MV _{ic}	Concern about risks/effects of classroom IT	3.45	0.91
	My professional development has benefitted	3.69	0.85
	My classroom practice has been transformed	3.70	0.72
	View of the role of IT in the classroom	4.37	0.94
MV _{gc}	We regularly review our classroom IT	3.24	0.99
	Classroom IT has increased sense of purpose	3.49	0.83
	Level of enthusiasm for classroom IT	3.82	0.88

Table 5.12 – Moderating and Mediating variables $n=1059$

5.7 Relationships between Vision, Leadership and Outcomes

The preceding sections set out the methodology by which each of the variables has been derived, whilst providing a detailed exploration of the data pertaining to each.

Table 5.13 summarises these variables, which will form the basis of the final analysis model and encapsulates the research framework described earlier in Figure 3.2.

Type	Locus	Variables
Independent	Leadership Paradigm	LP (Managerial, Participative, Instructional, Transformational)
Independent	Vision	Vt - Overall strength of technology vision Vr - Responsibility for the vision
Moderating & Mediating	Intermediate influence	MV _{it} – IT Competence MV _{ic} – IT Individual culture MV _{gc} – IT School culture
Dependent	Outcomes	O _e - Education related outcomes O _t - Teacher related outcomes O _s - School related outcomes

Table 5.13 – Overview of project variables

The number of variables and potential for complex interactions suggest that a multivariate analysis using sophisticated techniques is required. However, this is beyond the author's present ability in the time available for the project, either longhand in Excel or through acquiring and learning an appropriate statistical package (e.g. LISREL).

So as an alternative, a multi-stage approach was devised which sub-divides the analysis into a number of discrete and more easily manageable steps, as shown in Table 5.14, which results in a hybrid blend of direct and indirect effect models.

Stage	Activity
1	Analyse the relationship between the two independent variables (Leadership & Vision) to look for evidence of a correlation, such that these two variables might be expressed by a single 'Independent Variable' (IV) factor.
2	Analyse the relationships between this IV factor and each of the Dependent Variables, noting results and possible conclusions.
3	Analyse forward relationships between the intermediate variables (MV) and the Dependent Variables, noting possible correlations and effects.
4	Analyse backward relationships between the intermediate variables (MV) and the IV factor, noting possible correlations and effects.

Table 5.14 – Hybrid multi-stage analysis process

In all cases, the analyses in this final section are derived from calculations using raw or aggregated survey data, as opposed to re-using transformed or processed data constructs arising from other analyses already presented earlier in Section 5. The scale for most analyses reflects the Likert responses of Strongly Disagree (1) through to Strongly Agree (5).

Figure 5.18 illustrates each of the stages to be described, with arrow connections representing each variable regression phase, noting the descriptive paragraphs relating to each.

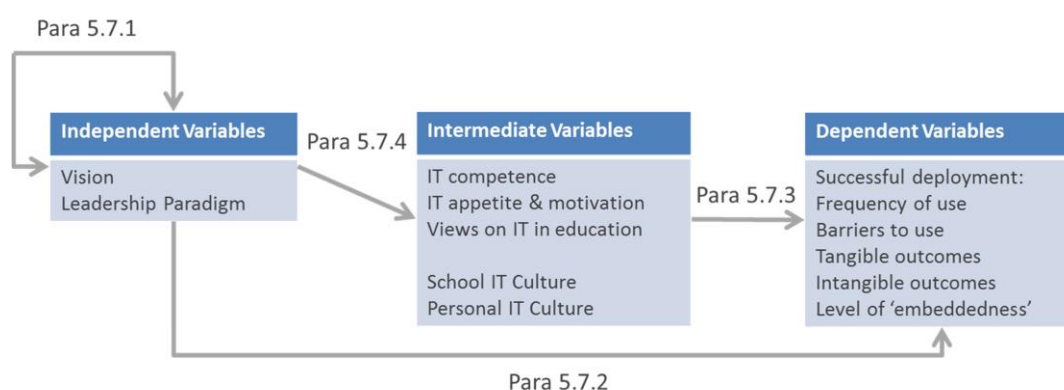


Figure 5.18 – Stages in testing the relationships between variables

5.7.1 Stage 1 – Relationship Between Leadership and Vision

A simple comparison of the distribution of the two independent variables for Vision (Vt and Vr) correlated against each of the leadership paradigms demonstrates a superficially good fit between both the Vt (Technology) and Vr (Responsibility) vision variables across leadership paradigms.

Leadership Paradigm	Vision	n (%)	Mean	SD
Instructional	Vt	195 (18%)	3.83	0.53
	Vr		3.46	0.80
Participative	Vt	320 (30%)	3.82	0.53
	Vr		3.19	0.90
Managerial	Vt	201(19%)	3.73	0.51
	Vr		3.21	0.89
Transformational	Vt	359(33%)	3.97	0.49
	Vr		3.08	0.93
		1059(100%)		

Table 5.15 – Correlation between Vision and Leadership $n=1059$

However, the graphical analysis (Annex 8) reveals underlying complexities which are not apparent in the summary data of Table 5.15. Whilst the *technology* related aspects of vision (Vt) do share broadly similar distributions across paradigms, the Managerial style is significantly less positively associated, indicating that vision is a weaker component of this leadership model.

Figure A6.2 illustrates the distribution for the *responsibility* aspects of vision (Vr), confirming the earlier indications in Figure 5.2 that some aspects of vision are not considered to be solely the responsibility of leadership, and likely to be shared or delegated to staff.

The graphical analyses show that this divergence of opinion maps well onto the different leadership paradigms, and hence, it is not necessary to regress separately for Vr and Vt across all leadership styles in subsequent analyses. In effect, the leadership paradigms can function as a factored variable for both leadership and vision.

5.7.2 Stage 2 – Relationship between leadership and dependent variables

Analysing leadership against the three factored measures of outcomes (Dependent Variables) reveals the *possibility* of a direct relationship;

Leadership Paradigm	Outcome	n (%)	Mean	SD
Instructional	Oe	195 (18%)	3.34	0.37
	Os		3.48	0.56
	Ot		3.70	0.66
Participative	Oe	320 (30%)	3.29	0.44
	Os		3.48	0.57
	Ot		3.62	0.68
Managerial	Oe	201(19%)	3.32	0.41
	Os		3.43	0.56
	Ot		3.61	0.67
Transformational	Oe	359(33%)	3.39	0.39
	Os		3.60	0.57
	Ot		3.80	0.60
		1059(100%)		

Table 5.16 – Correlation between Independent and Dependent Variables $n=1059$

Benefits to the school (in terms of the general learning environment) emerge as the most widely attributed value of classroom technology projects, with positive outcomes for teachers being ranked second. Tangible educational benefits for students are surprisingly ranked as the least significant outcome in the study.

These opinions are generally similar across all leadership paradigms, though from the analysis it is clear that Managerial and Transformational styles may have a greater positive impact than Instructional and Participative.

We can conclude from this that there is *some* evidence of an effect between leadership style and project outcomes, though there is insufficient detail at this stage to properly understand the mechanisms involved.

5.7.3 Stage 3 – Relationship Between Intermediate and Dependent Variables

The preceding analyses (paras 5.7.1 and 5.7.2) have focussed on data which might be indicative of a *direct effect model* relationship between the Independent and Dependent variables. To explore the possibility of more nuanced, *indirect effect model* relationships we must firstly analyse possible correlations between the Intermediate variables (Moderating and Mediating) and the Dependent variables.

A one-way ANOVA for each of the dependent variables shows significant variances and very low p-values. Clearly, the intermediate variables (Moderating and Mediating) *do* influence the Dependent variables; characteristics of technology, school culture and individual culture introduce a significant variable effect on outcomes, as measured in terms of educational benefits, school benefits and teacher benefits.

Variable	Mean	SD	Var	p
Oe	3.34	0.41	0.16	5E-82
Osc	3.50	0.57	0.32	3E-37
Ot	3.69	0.65	0.43	1E-26

Table 5.17 – One-way ANOVA, Moderating and Mediating variables correlated against Dependent Variables $n=1059$

This is an important finding. Having previously found little compellingly clear evidence of a *direct effect* link between independent and dependent variables, these results show strong indications of an *indirect effect* connection. A series of graphs depicting the relationships of individual dependent variables to the intermediate variables is presented in Annex 9.

5.7.4 Stage 4 – Relationship Between Leadership and Intermediate Variables

Testing the hypothesis of indirect effects through a one-way ANOVA analysis delivers the results given in Table 5.18 below.

Leadership Paradigm	Outcome	n (%)	Mean	SD	Var
Instructional	MVit	195 (18%)	3.60	0.57	0.33
	MVic		3.47	0.68	0.45
	MVsc		3.56	0.43	0.18
Participative	MVit	320 (30%)	3.63	0.58	0.33
	MVic		3.44	0.69	0.48
	MVsc		3.57	0.43	0.19
Managerial	MVit	201(19%)	3.62	0.60	0.36
	MVic		3.37	0.67	0.45
	MVsc		3.58	0.43	0.19
Transformational	MVit	359(33%)	3.62	0.57	0.32
	MVic		3.68	0.66	0.44
	MVsc		3.60	0.40	1.16
		1059(100%)			

Table 5.18 – Correlation between Leadership and Intermediate Variables $n=1059$

From the data, we can conclude that the leadership paradigm *does* induce different effects on the intermediate variables. Detailed graphs illustrating these data are presented in Annex 10, with Figure 5.19 below providing a triangulated summary. It is clear that the *Transformational* leadership paradigm is most closely correlated to positive outcomes than any other, with Managerial having the least beneficial effect.

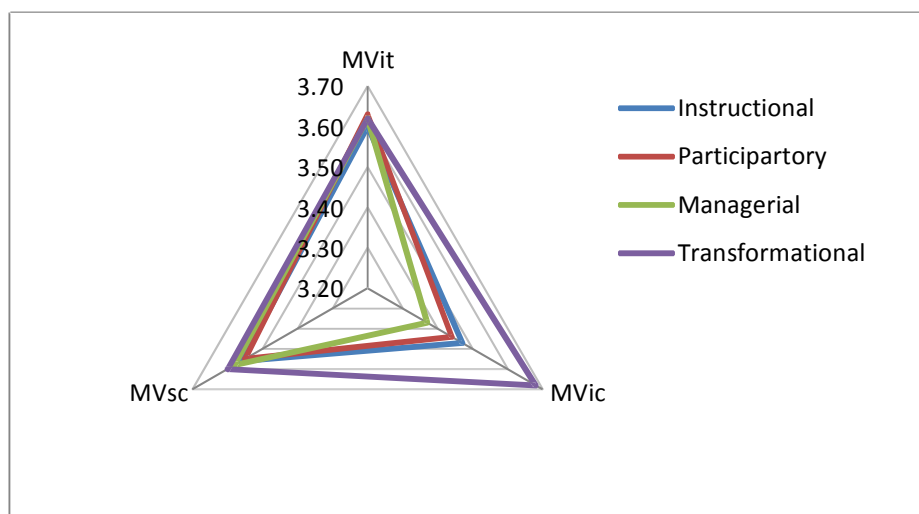


Figure 5.19 – Correlation between Leadership paradigms and independent variables, $n=1059$

In particular, the impact of leadership style on individual teacher culture shows significant variation, and is considerably more positively affected by the transformational paradigm than other styles.

6 Research Conclusions

This has become a *much* bigger project than the author had anticipated, resulting in a very large dataset from which a substantial number of variables, inter-related dependencies and effects might be found and extrapolated. Section 5 attempted to explore the most important and relevant findings, though the underlying data offers abundantly more analysis potential than has been exposed so far.

In these conclusions, only the principal findings will be discussed.

6.1 Vision

The analysis results relating to vision are in some ways surprising and paradoxical. Overall, the findings support the premise that vision is an important component of leadership, and that those leadership paradigms in which the role of vision is more strongly characterised have a more positive correlation with improved outcomes in technology projects.

Summarising the key points, we find that:

Vision is generally felt to be everyone's business but the leadership's responsibility;

Most respondents claim that vision is a product of the schools internal processes, whilst at the same time acknowledging the considerable impact of external influences;

Fear of being left behind (reactive) is a stronger motivation than creativity and innovation (proactive);

Retrospective and future visions of classroom technology are positive in the majority of cases, with the majority agreeing that a clear vision was in place for previous projects and look forward to new technology projects.

The characterisation of vision across the leadership paradigms varies, with Transformational and Instructional models being positively correlated and the Managerial style being less strongly correlated to outcomes.

The associations between the technology related aspects of vision show a good consistency of opinion across leadership styles, inferring that irrespective of the leadership style there is a positive technology vision in the majority of UK schools.

However, opinions regarding the responsibility aspects of vision vary markedly, and there is a considerable shift of ownership between leadership styles.

The Managerial and Instructional styles tend to regard vision as a predominantly leadership responsibility, whereas Transformational and Participative are more likely to feel that the vision should at least be shared, perhaps delegated.

The findings are consistent with the visionary leadership models described by Nanus (1992) and suggest that there is good application of the vision-building behaviours espoused by Leithwood et al (1997).

6.2 Leadership

The four leadership paradigms selected for this analysis were an attempt to blend, from a comprehensive study of the literature, multiple aspects of leadership typology. These included traditional and contemporary, centralised and decentralised, task and people, bureaucratic and visionary.

In some ways this was a risky approach, as the four models do not comprehensively exemplify all paradigms described in the general literature, and do not map precisely onto the full range of leadership characteristics studied in recent years. However, to attempt to embrace all possibilities would have been unfeasible; the research would have been unmanageable and the analysis pointlessly complex. The simplification to four models, studying the concepts of vision and leadership in relation to the 2x2 framework defined in Figure 2.5 supported a focussed yet comprehensive exploration of the most important characteristics of educational leadership. Importantly, these leadership paradigms closely correlate with the approach similar educational research, such as Leithwood et al (2012)

Summarising the key points of the research, we find that:

Leaders with characteristics which are representative of the Transformational model are the most commonly described style in the research. Followed closely by Participative, and with Managerial and Instructional being in third and fourth place;

Leadership style *does* have an impact on technology project outcomes, though this is predominantly an indirect effect, arising from the influence of leaders on school and teacher culture;

Leadership style appears to have a lower impact on the intermediate variables associated with technology implementations, with results for all paradigms being consistently distributed;

The four leadership paradigms forming this research may not have encapsulated every aspect of leadership typology, but nevertheless, they clearly capture a significantly diverse range of aspects and characteristics relevant to education to have exposed a number of key findings.

6.3 Quantifying the Differences in Leadership Style

Bringing this all together, we can conclude that vision and leadership *do* have a substantial impact on the effectiveness of classroom technology projects, when evaluated across several distinct measures.

To simplify understanding of the analysis results, the main findings can be expressed as an effect size, which provide a sense of the relative impact of each leadership paradigm.

Leadership Paradigm	Intermediate Variables			Dependent Variables		
	MVit	MVic	MVsc	Oe	Os	Ot
Managerial	-	-	-	-	-	-
Instructional	-0.017	0.021	-0.023	0.020	0.044	0.067
Participative	0.058	0.051	0.064	0.035	0.027	0.038
Transformational	0.064	0.243	0.124	0.087	0.122	0.146

Table 6.1 – Effect Analysis, Leadership Style to Intermediate and Dependent Variables
n=1059

This effect size analysis takes the Managerial model (overall, the worst performing) as the base line, and calculates the relative impacts that each of the other leadership paradigms might achieve.

The impact of leadership style on the intermediate variables changes considerably, with the Transformational model in particular having a 24% positive benefit on individual teacher culture ratings than under the base Managerial leadership style, and a 12% gain for school culture.

These of course are intermediate variables, not outcomes, and we find that opinions of the overall impact of the transformational style are slightly weaker when measured at the dependent variable stage, with improvements of 12% and 14% respectively for school and teacher related improvements. Clearly there are additional moderating factors at work here, perhaps amongst the Antecedent variables of Table 3.2, which would require further analysis or perhaps another research project to more fully understand.

Most importantly we can see an 8% improvement in opinions of educational outcomes related to the transformational paradigm, though clearly, this finding needs to be externally cross-referenced against actual academic results such as SATs or OfSTED to properly confirm the measurable achievement benefits.

Benefits on this scale are consistent with the findings of other research into the gains arising from classroom technologies, such as Marzano (2009), noting a 17% gain, and EEF (Education Endowment Foundation, 2014) who, through a meta-study of twelve research projects concluded that effective classroom IT projects can yield four months of learning gains per year.

6.4 Implications for the Education Community

We have already seen that as far as the UK schools which have been the subject of this research are concerned, there is evidence of good vision, planning and preparation across all leadership paradigms. Ownership of the vision varies, as do the detailed effects on teacher and school culture, which in turn have downstream implications for overall outcomes.

However, it is clear that greater positive value can be derived from projects under the transformational leadership model than from any other, and school governors, head teachers and perhaps those at policy level must understand the detailed ramifications of developing transformational behaviours across all aspects of school leadership.

Whilst we can be pleased that the National College for Teaching and Leadership have focussed their training on this model for several years, we must also be concerned that those students in schools with a predominantly managerial style are potentially being disadvantaged by 12-14%, and there is much work still to do.

Also, we must note that whilst the participation rate of this research was very good and *potentially* representative of a large population, it's clear that response was entirely voluntary and therefore self-selection may have induced some bias in the dataset, in that respondents may already be favourably disposed to the topics of technology and its potential for transformational value.

6.5 Recommendations for further research

Although this research could well prove valuable and interesting as it stands, further refinement of the research process might improve the integrity and quality of the data. For example, specifically selecting a sample of schools fully representative of type, area, student demographics and so on might yield a more representative dataset than the largely self-selected dataset which currently forms the basis for this research.

More generally, this research has focussed solely on UK schools, which are likely to have specific characteristics which mean that these findings only apply to the UK, and should not be assumed to be relevant in other countries, developed or developing. Specific research in each context is recommended.

7 Reflective Assessment*

As the author's first ever research project, this work has been something of a challenge. The advice of the YSJ and RKC staff to select a topic with a narrow focus is now much more fully understood, as with hindsight, the scale of this project might have been dramatically reduced. The high response rate to the questionnaire did not create any specific problems. Though on reflection, there were too many questions giving rise to too many variables, too many permutations and a mind-twisting series of statistical models to construct, manipulate and assimilate.

Despite these reservations, the research has provided a valuable opportunity to study a topic of interest in considerably greater breadth and depth than might otherwise have been possible. Having a *tangible* project around which to structure literature searches, study related educational research projects, design and execute a large scale survey, then analyse and summarise findings has proved hugely rewarding. Both in terms of the outcomes of the project itself, and in the substantial learning opportunities afforded to the author.

Whilst this paragraph might mark the end of the dissertation, it does not, quite, mark the end of the project. There is still the promised research summary to be prepared and despatched to those questionnaire respondents who expressed an interest, and still much useful data not yet fully analysed in the research dataset which might yield further significant findings.

And then there are those future projects, which hopefully, this work will benefit...

***See also the final pages, for a reflective assessment written after the dissertation had been submitted, addressed to fellow and future MALIC colleagues**

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Annex 1 – Early models of Educational Leadership

Model	Scientific Management	Bureaucracy	Collegial	Political
Generative Metaphor	Rational	Mechanistic	Organic	Bargaining
Basic Unit of Analysis	Engineering, Computer Programming, Assembly Line, Dehumanizing	Factory, Machine, Blue Print, Legal Code	Health, Missionary, Plant, Nurturing	Coalition, Trade-off, Game, Dealing
Major Concepts	Goals and objectives, Alternatives, Cost-benefit analysis, Maximizing decisions, Accountability	Organisational routines implicit in structure and function, role and rules, and rights and prerogatives	Commitment of supportive work group and school goals	Individuals and organization as players and actors, Rules of the game, Stakes and interests
Authority Type	Reward, Coercive	Reward, Coercive, Legal position	Normative	Instrumental
Administrative Prescription	Plan, organize, control, evaluate	Emphasize, Formalization, Centralization, Stratification, Efficiency, Production in organization structure	Use team administration and power-equalization strategies to develop a highly effective work group committed to school	Assume brokerage role to build coalition among competitive and conflicting interests within the school and between the school and it's environment
Principal as Educational Leader	High	Low	Medium	Medium
Principal as Manager	High	High	Low	Medium
Consequences of the Principals role	The principal is in complete control but teacher commitment is low and alienation is high. Public participation is low.	Educational leadership is implicit in the standard curriculum and in student operating procedure rather than in principals and teachers. Teachers are often 'satisfied' but unchallenged. Commitment is lethargic. Public participation is low. The organization is in control.	Expressions of educational leadership are widespread and teachers are influential but schooling is expensive and accountability is low. Professionals are in control.	Wider participation by all groups but since authority is widely distributed; decisions are often made on the basis of compromise. What is acceptable rather than what is best becomes the decision role. Control shifts with issues and with coalitional strengths.

Table A1.1 Early models of Educational Leadership (Sergiovani, 1979)

Annex 2 – Selecting the Four Leadership Paradigms

Scholars of leadership will quickly discover that there are many descriptions of the concept, and that these have evolved considerably over time. This adds significantly to the complexity of any research which includes leadership as one of its main themes. This is equally true of generalist definitions of leadership, and also of the specialist definitions in more constrained contexts such as the educational sector.

Leithwood et al (2012) studied 121 “feature-length” articles in education journals from the UK, North America, Canada, Australia and New Zealand going back as far as 1988, in order to gain a comprehensive understanding of studies of educational leadership models. They identified twenty distinct concepts, with instructional and transformational being most frequently mentioned. The next most frequently mentioned concepts were moral leadership, managerial leadership and cultural leadership.

Leithwood and his team identified six main leadership typologies, clustering together the main characteristics of the twenty models identified from the literature. These are set out in Table A2.1 below, accompanied by the main characterisations defined by Leithwood.

For this research, Leithwoods six leadership paradigms have been further reduced to four for the purposes of this dissertation, using the following logic:

Moral leadership shares many similar attributes to transformational leadership. Indeed, Burns (1978) strongly connects the motivational aspects of leadership with higher moral positioning, and his work firmly connects transformational leadership with moral leadership. So whilst accepting the possibility that moral leadership might exist as a separately defined concept, for the purposes of this research we will assume that moral leadership is subsumed into transformational leadership.

Contingent leadership emphasises that effective leaders adapt their style according to context. Hence, it could be argued that it is not a distinct paradigm, but rather, that astute leaders choose the appropriate model according to the present need. For the purposes of this research, we will assume that any contingent leaders will have selected one of the four paradigms being studied.

Accordingly, Leithwoods original six ‘composite’ models are further reduced, to include only Transformational, Participative, Instructional and Managerial.

Approaches to School Leadership	Who exerts influence	Sources of influence	Purposes for influence	Outcomes of influence
Instructional	Typically those in formal leadership roles, essentially school principals	Expert knowledge Positional power	Enhance the effectiveness of teachers classroom practices	Increased student growth
Transformational	Typically those in formal leadership roles, but not always restricted to those persons	Inspire higher levels of commitment and capacity among organisation members	Greater effort and productivity Develop more skilled practice	Increased capacity to continuously improve
Moral	Those in formal administrative roles	Use of a system of moral values to guide organisational decision making	Increase sensitivity to the rightness of decisions Increase participation in decisions	Morally justified courses of action Democratic schools
Participative	The group (including non-administrative members)	Interpersonal communication	Increase participation in decisions	Increased capacity of organisation to respond productively to internal and external demands More democratic organisation
Managerial	Those in formal administrative roles	Positional power Policies and procedures	Ensure efficient completion of specified tasks by organisation members	Achieve formal goals of the organisation
Contingent	Typically those in formal roles	Matching behaviour to organisation context Expert problem-solving processes	Better meet needs of organisation members More effective responses to organisations challenges	Achieve formal goals of the organisation Increased capacity of organisation to change.

Table A2.1 – Characteristics of Educational Leadership Models (Leithwood et al, 2012)

Annex 3 – Questionnaire Piloting & Response Rates

The structure and progress of the surveys are summarised in Tables A3.1 to A3.3 below, detailing the pilots and the ‘live’ surveys in terms of the number of sub-sections at each stage, the number of invitations response rates and implications.

Survey Name	Invitations Sent	Total Responses	Response Rate	Comment
Pilot 1 Initial	460	5	1%	Poor
Pilot 1 Reminder	460	1	0.2%	Disastrous
Pilot 2 Initial	460	2	0.4%	Decision to try Survey Monkey!

Table A3.1 – Response rates to pilot Questionnaire invitations delivered via QuickSurveys

Survey Name	Invitations Sent	Total Responses	Response Rate	Comment
Pilot 3 Initial	442	20	4.5%	Much better – can work with this.
Pilot 3 Reminder	437	6	1.4%	Takes Pilot 3 total to almost 6%. Good.
Pilot 4 Initial	451	13	2.9%	Sent on a Friday – low response cause?
Pilot 4 Reminder	449	15	3.3%	Sent on Wednesday. Total Pilot 4 6.2%
Pilot 5 Initial	453	22	4.8%	Refined cover letter style
Pilot 5 Reminder	438	11	2.5%	Total 7.3% response rate for Pilot 5

Table A3.2 – Response rates to pilot Questionnaire invitations delivered via SurveyMonkey

Survey Name	Invitations Sent	Total Responses	Response Rate	Comment
Live 1 Initial	4016	194	4.8%	
Live 1 Reminder	3948	83	2.1%	Total for Live 1 - 6.9%
Live 2 Initial	4050	225	5.5%	
Live 2 Reminder	3962	99	2.5%	Total for Live 2 - 8%
Live 3 Initial	3985	205	5.1%	
Live 3 Reminder	3917	96	2.5%	Total for Live 3 - 7.6%
Live 4 Initial	4002	228	5.7%	
Live 4 Reminder	3944	107	2.7%	Total for Live 4 - 8.4%
Live 5 Initial	3994	217	5.4%	
Live 5 Reminder	3918	104	2.7%	Total for Live 5 - 8.1%
Live 6	39	4	10%	

Table A3.3 – Response rates to ‘live’ Questionnaire invitations delivered via SurveyMonkey

Notes:

SurveyMonkey pre-checks email addresses and blocks any that have previously opted-out from future surveys, hence the variable (and reducing) number of invitations sent.

Relatively small variance between response rates to the 5 mailing lists, suggests a good random distribution throughout the email lists.

‘Live 6’ (39 participants) represents re-sent invitations to new email addresses gleaned from ‘auto-responses’ to the original invitations, indicating an address change.

Annex 4 – Ethical Factors and Associated Measures

Factor	Design & Implementation Measures
Verifying Identity	This is an anonymous survey, where it is not necessary to confirm that the responses are from an intended participant. Most of the email addresses are generic such as 'admin@..' and 'office@..', though in some cases the email addresses do name the respondent. From a research perspective, only the role of the participant is relevant, not the identity. In any case, all personal identity information is deleted from the dataset (see below).
Public/private space	All email addresses are those made publicly available by the schools themselves, and should not therefore present an issue of private space invasion.
Informed Consent	An inherent aspect of Survey Monkey is that all surveys give the recipient the opportunity to opt-out of future surveys, so email address lists presented to Survey Monkey are automatically trimmed to remove any previously opted-out addresses. Furthermore, the cover letter made it clear that participation was 'kindly requested' and therefore optional. Recipients of the initial invitation mail were given the opportunity opt-out of not just the present survey, but also to receiving a follow-up reminder the following week, and any future Survey Monkey requests from any source.
Levels of control	A difficult issue; the nature of this kind of internet mediated research makes it difficult to control who will respond, under what environmental conditions (relaxed, stressed, etc), the equipment used, the time taken, multiple responses from the same individual, etc, etc. Survey Monkey does provide good levels of built-in management of some of these issues, and the responses themselves contain other 'data forensics' on time taken, etc, to assist in analysis management.
Withdrawal	The cover letters made it clear that the survey is completely optional. Furthermore, all partially completed surveys have been excluded from the analysis, as non-completion is regarded as inferred withdrawal.
Debriefing	Individual debriefing is not feasible given the scale, and not appropriate given the anonymous nature of the research. All participants were offered the opportunity to receive a summary of the research, providing a detailed level of debriefing on the research findings.
Deception	There are no attempts to deceive participants, either in purpose of the survey, the way in which data would be used, or in any other aspect of the research and its analysis. Efforts were made at all times for communications to be simple and clear, to avoid possible ambiguities and confusion, and hence, the possibility of 'unintended deception'.
Monitoring & Participant Protection	Monitoring is not feasible in this form of internet mediated research; however, there is nothing in the design of the survey which is likely to induce stress or anxiety in participants.
Data Protection	Any information which could identify the source of each response was deleted immediately from the original dataset received from Survey Monkey. This included those email addresses provided by participants requesting a copy of the research summary. These email addresses were kept in separate file, and could not be re-associated with survey data.

Table A4.1 – Ethical Factors

Annex 5 – Questionnaire Cover Letters (Email)

From: ukma.student@gmail.com
Subject: Would you help me with my MA research (it'll take <10mins)?

Hi,

I'm a mature student currently undertaking a Master's degree in "Leading Innovation and Change". My dissertation is researching the processes of change relating to the introduction of classroom technology such as interactive whiteboards, projectors, tablets, and so on.

I've created a short survey, the data from which will form part of the primary research for my dissertation.

It'll take around 10 minutes and I'd be really grateful if you could find the time to complete it, the closing date is one week from today. It would be extremely helpful if you could also forward this mail to colleagues in your school, so I can get the widest range of views.

You can access the survey here: [SurveyLink]

It's completely anonymous; the survey won't ask you for personal or school ID info.

Thanks for your help!

Peter.

PS. If you'd like to get a summary of the research findings (so you can find out what lots of other teachers think!) you'll be able to enter an email address at the end of the survey. It's optional, and please be assured that your email address will be kept separate from the research data and will be deleted once I've forwarded you the report.

I'll probably send out gentle reminders just before I need to close the survey, if it will annoy you to receive this just click here: [RemoveLink]

Figure A5.1 – Initial invitation email format

From: ukma.student@gmail.com

Subject: Would you help me with my MA research (it'll take <10mins)?

Hi,

I hope you don't mind me sending you a gentle reminder about my survey – a big “thank you” to those who have already participated!

The majority of respondents so far are Head Teachers, so to get a full range of views I'd be grateful if you might forward this mail to colleagues throughout your school. I'll leave the survey open for a few more days to give everyone time to participate.

A recap on the context.....

I'm a mature student currently undertaking a Master's degree in “Leading Innovation and Change”. My dissertation is researching the processes of change and leadership relating to the introduction of classroom technology such as interactive whiteboards, projectors, tablets, and so on.

I've created a short survey to gather opinions on these matters, which will form part of the primary research for my dissertation.

It'll take around 5-10 minutes and I'd be really grateful if you could find the time to complete it within the next day or so.

You can access the survey here: [SurveyLink]

It's completely anonymous; the survey won't ask you for personal or school ID info.

Thanks for your help!

Peter.

PS. If you'd like to get a summary of the research findings (so you can find out what lots of other teachers think!) you'll be able to enter an email address at the end of the survey. It's optional, and please be assured that your email address will be kept separate from the research data and will be deleted once I've forwarded you the report.

Depending on how the survey goes, I might send out a very short follow-up survey to get some clarifications on points I missed the first time around (I'm new to this!). If it will annoy you to receive this just click here: [RemoveLink]

Figure A5.2 – Reminder invitation email format

Annex 6 – Questionnaire Design – Data Requirements Table

The following tables summarise the questionnaire design in terms of:

Overall research question

Type of research

Question sub-categories

The ‘data requirement objective’ for each question (ie what aspect of the research each question seeks to illuminate, or the overall purpose for the question)

The question type, which is one of:

- AT. – Attribute question
- OP. – Opinion question
- RA. – Rating question
- LI. – List question

The ‘class’ of question, which indicates how the data will be classified and codified for analysis purposes, which is one of:

- Cat. – Category; enables respondent data to be categorised (eg, by length of service, role type, etc)
- Tech. – Technology; provides information regarding technology utilisation and effectiveness
- Lead. – Leadership; provides information regarding school leadership
- Vis. – Vision; provides information regarding overall vision
- Surv. – Survey; provides information and feedback on the survey process itself.

Research Question/Objective: Null Hypothesis: Leadership and Vision are not significant factors in the effectiveness of classroom technology projects in the UK				
Type of Research: Largely descriptive (attitude & opinion), with the potential for explanatory/analytical assessment.				
Question sub-category 1: Information about the respondent.				
Question	Options	Data Requirement Objective	Type	Class
How many years have you been teaching?	Free text numeric entry	Will enable analysis of experience, with additional potential to investigate correlation of pedagogical preferences - 'traditional' vs 'progressive' models.	AT.	Cat.
Which of the following best describes your role?	Primary head teacher/deputy head/assistant head Primary subject co-ordinator Primary AST Primary classroom teacher Secondary head teacher/deputy head/assistant head Secondary head of faculty/head of department Secondary AST Secondary classroom teacher HE Teacher/Lecturer Administrator Other (please specify)	To allow i) responses from 'non-educational' participants to be excluded, and ii) facilitate comparisons between different types of respondent.	AT.	Cat.

How would you rate your IT skills?	Very IT literate, able to deal with pretty much anything. Comfortable with IT, but needing help and support with new stuff. Somewhat fazed by IT, concerned that it can be complex, waste time and even be disruptive.	To assess general IT capability of the respondent, and hence, their likely competence and appetite for classroom technology generally.	OP.	Tech.
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Question sub-category 2: Information about the technology.				
Question	Options	Data Requirement Objective	Type	Class
Which of these best describes the technology you have in mind during this survey?	Individual student laptop/PCs Individual student iPad/tablets Teacher PC + Projector Teacher PC + Interactive Projector Interactive Whiteboard Other	Classroom technologies vary in their complexity, application (in terms of pedagogy) support needs, novelty (IWB's = 10-15yrs, iPads 0-4yrs). The response here will enable data to be analysed by technology type, and may also enable inferences of technology competence and appetite to be drawn.	AT.	Cat. Tech.
If you chose Interactive Whiteboard or Interactive Projector, would you say that you:	Fully exploit the interactive functions Occasionally use the interactive functions Never use the interactive functions – just use it like a projector and screen Not Applicable	To assess the extent to which the more sophisticated aspects of the technology are used, indicating competence and appetite.	AP.	Cat. Tech.

With this classroom technology, would you say that you:	Fully exploit it to the maximum potential, in every lesson Make use of it regularly Rarely use it If you chose 'rarely use it', is this mainly because: It doesn't work reliably It doesn't fit well with your subject or teaching style You find it hard to get good e-lesson materials You haven't had enough training or support Other (please specify)	To assess the frequency with which the technology is used, indicating competence and appetite.	OP.	Cat. Tech.
Would you like to see:	Significantly increased use of classroom technologies Slightly increased use of classroom technologies About the same levels of technology use as we have today Less technology in classrooms	To assess the appetite for further expansion of the role of classroom technology, as an indication of competence and also of successful current experiences. (Positive experience are likely to yield an appetite for more technology. Negative experiences for less).	OP.	Cat. Tech.
The greatest challenge to introducing new classroom technologies in future is:	Leadership and change management Finding new 'teaching & learning models' to fully incorporate technology Constantly changing technology Training and support Other (please specify)	Respondents are likely to consider future technology deployments based of the opinions of previous & current experiences of technology projects. This 'top level' question seeks to gather a general guiding response of where respondents feel that the key issues lie – with technology, vision, leadership, change management.	OP.	Cat. Tech. Lead. Vis.

Which additional technologies are you likely to purchase in the next 12-18 months (tick all that apply)?	Individual student laptop/PCs Individual student iPad/tablets Projector Interactive Projector Interactive Whiteboard System Interactive LCD (large, on wall) Other (please specify)	Similar to the historical/current question, we seek here a forward view on types of technology as an indication of progressive thinking (IWBs/Projectors = more traditional, ipads/tablets = more progressive)	OP.	Cat. Tech.
The greatest challenge to introducing new classroom technologies in future is:	Leadership and change management Finding new 'teaching & learning models' to fully incorporate technology Constantly changing technology Training and support Other (please specify)	Similar to the historical/current question, we seek here a forward view on anticipated challenges. This is partially a 'check' of the earlier question, but also encourages respondents to think more specifically about new projects rather than existing/historical ones.	OP.	Cat. Tech. Lead. Vis.

Question sub-category 3: Strategy and Vision.				
Question	Options	Data Requirement Objective	Type	Class
We felt there was a very clear and compelling national educational vision supporting the decision to implement this technology We felt there was a very clear and compelling school level educational vision supporting the decision to implement this technology All staff were fully consulted when decisions to implement these new technologies were taken There was widely accepted consensus that introducing this technology represented an important and necessary change	Likert Scale (5 ratings – Strongly Agree to Strongly Disagree)	Investigating the connection to a general educational policy, developments and activities in other schools. (ie extrinsic vision and leadership).	RA.	Vis. Lead.
We mainly introduced this new technology because...	We regard ourselves as educational innovators and risk takers The technology benefits were considered well-proven, so no reason not to We didn't want to feel that our school was being left behind Funding had been made available and we wanted to spend it	Indication of leadership style and school vision/culture. Progressive/traditional, Proactive/reactive, Complex/simplistic.	OP.	Lead. Vis.

Question sub-category 4: Project Management and Implementation				
Question	Options	Data Requirement Objective	Type	Class
Manage the practical and technical aspects of project Manage the training and teacher support aspects of project Manage communications; to staff, students, parents Ensure that adequate new e-learning materials would be available Ensure that the objectives of the initial vision would be fully met	Likert Scale (5 ratings – Strongly Agree to Strongly Disagree)	Understand whether respondents felt that this aspect of the project could have been handled better	RA.	Lead.

Question sub-category 5: Change Management				
Question	Options	Data Requirement Objective	Type	Class
The majority of staff were extremely enthusiastic about the plan to introduce these technologies The planning and preparation for this project was a cause of conflict between staff, or between staff and school leadership No major risks or concerns were highlighted by staff during the planning stage Any major risks or concerns were thoroughly discussed and accommodated	Likert Scale (5 ratings – Strongly Agree to Strongly Disagree)	Understand whether respondents felt that this aspect of the project could have been handled better	RA.	Lead.

Question sub-category 6: Leadership & Vision				
Question	Options	Data Requirement Objective	Type	Class
We felt there was very clear and visionary leadership on this project We had a specific project team to work on the project We had a clear sense of 'shared mission' right from the start Actually, the project has strengthened us as a team, bringing us closer together The project has helped to further develop a sense of 'shared educational Objective' in our school	Likert Scale (5 ratings – Strongly Agree to Strongly Disagree)	Understand the extent to which respondents felt there was a clear vision, effective leadership and cohesive purpose	RA.	Lead.

Question sub-category 7: Impact of the Technology/Project				
Question	Options	Data Requirement Objective	Type	Class
This new technology has had a transformational, positive impact on student learning in my classroom Personally, I've learned a lot and the technology has had a transformational, positive impact on my classroom practice and personal development Actually, I don't feel that the new technology has had a transformational impact, it's really just an incremental change to existing classroom practice It would be very difficult to return to a 'traditional' teaching model and have to manage without this technology It's possible to directly correlate improvements in student achievement to the introduction of this technology The main benefits of this technology are in the positive impact on school/classroom practices rather than on student achievement We have routine reviews of the value of this technology and feed this back into our strategic thinking Generally speaking, everything went according to plan and we are achieving the outcomes we'd planned for	Likert Scale (5 ratings – Strongly Agree to Strongly Disagree)	Understand how respondents felt the project has changed their classroom practice, their pedagogical skills, overall perceptions of outcomes, and whether the impact is quantifiable.	RA.	Lead.

Question sub-category 8: School Culture				
Question	Options	Data Requirement Objective	Type	Class
Our school culture: Provides excellent mentoring and support Is strongly focussed on results at all costs Supports teamwork, consensus, and participation Fosters trust, openness and participation Values the provision of new resources and creating new challenges Encourages staff to be creative and try out new things, even if this is outside of accepted norms Is a very controlled and well-structured environment. Has lots of formal procedures which staff must follow	Likert Scale (5 ratings – Strongly Agree to Strongly Disagree)	Proxies for ‘direct questions’ on leadership - from responses to these questions the objective is to extrapolate characteristics which might infer the dominant leadership style(s) in the school.	RA.	Lead.

Question sub-category 9: School Leadership				
Question	Options	Data Requirement Objective	Type	Class
In my school, responsibility for the following activities is the MAIN responsibility of: Managing timetables and staff issues Setting achievement goals and expectations Major decisions about curriculum development Building school vision and mission Creating a productive school culture Planning and budgeting Policy regarding technology and innovation Monitoring curriculum and instruction Ensuring stability and consistency Promoting and defending core educational values Mentoring, coaching and staff development Promoting and defending moral and ethical standards Creating a positive, motivational and engaging learning culture	Governors, Head Teacher, Heads of Dept., Classroom Teachers, School Manager, Admin staff, Everybody, Nobody/External	To understand who in the school has the primary responsibility for the specified activity, role, or behaviour.	RA.	Lead.

Question sub-category 9: School Leadership				
Question	Options	Data Requirement Objective	Type	Class
The leadership model/style in my school is: Strongly influenced/constrained by local community challenges Firmly the responsibility of the Head/Principal Distinctly separated from day to day management issues Mostly involved with preserving the school culture and ethos and representing the figure of authority Routinely involved in the practical detail of school management Primarily focussed on developing and promoting a school vision Predominantly about staff communication; guidance and feedback Constantly being adapted to meet short term demands Very focussed on managing external stakeholders and agencies Heavily geared towards achievement against SATS/OFSTED Organised around staff participation and shared decision making	Likert Scale (5 ratings – Strongly Agree to Strongly Disagree)	To understand whether each of the specified leadership behaviours should be fully attributed to the head teacher.	RA.	Lead.

Question sub-category 10: Survey Feedback				
Question	Options	Data Requirement Objective	Type	Class
Regarding the questions:	There were too many – I wouldn't have started if I'd known there would be this many questions! It was just about right. This is important stuff, I'd have been happy to spend another 5-10mins on it.	Initially intended for use during the pilot (to judge the respondents view of the whole survey), the response to this question could be an indicator of respondent's attitude to classroom technology generally. Especially those respondents who selected the third option, and would have been prepared to answer more questions based on a strong interest in the topic.	LI.	Surv.
Relevancy of the research/survey:	It was highly relevant – I was able to answer in detail about a specific project. It was generally relevant – I was able to give broad answers about classroom technologies in our school. It wasn't especially relevant – my answers are general opinions, but don't relate to a specific project.	Initially intended for use during the pilot (to judge the respondents view of the whole survey), the response to this question could be an indicator of respondent's attitude to classroom technology generally. Especially those respondents who selected the third option, where it is arguable that these responses should be excluded from the analysis. Additionally	LI.	Surv.
Direct Feedback:	If you have any extra comments on the topics in this survey, please feel free to jot down any points in the space below (optional)	Free text entry. Again intended for the pilot only, though this does provide respondents with an opportunity for general comment.		Surv.

Annex 7 – Typology of Leadership Characteristics, Direct and Inverse

Leadership Paradigm	Behaviour indicators with a direct relationship to the leadership style	Behaviour indicators with an inverse relationship to the leadership style
Transformational	Primarily focussed on developing and promoting a school vision. Distinctly separated from day to day management issues. Firmly the responsibility of the head/principal Mostly involved with preserving the school culture and ethos and representing the figure of authority. Major decisions about curriculum development. Building school vision and mission Creating a productive school culture. Creating a positive, motivational and engaging learning culture. Policy regarding IT and innovation	Routinely involved in the practical detail of school management. Constantly being adapted to meet short term demands. Planning and budgeting. Managing timetables and staff issues.
Instructional	Mentoring, coaching and staff development. Monitoring curriculum and instruction. Managing timetables and staff issues. Major decisions about curriculum development. Creating a productive school culture. Promoting and defending core educational values. Creating a positive, motivational and engaging learning culture. Values the provision of new resources and creating new challenges. Heavily geared towards achievement against SATS/OfSTED	Building school vision and mission Policy regarding IT and innovation Planning and budgeting. Strongly influenced/constrained by local community challenges

Leadership Paradigm	Behaviour indicators with a direct relationship to the leadership style	Behaviour indicators with an inverse relationship to the leadership style
Managerial	Routinely involved in the practical detail of school management. Constantly being adapted to meet short term demands. Very focussed on managing external stakeholders and agencies. Planning and budgeting. Managing timetables and staff issues. Firmly the responsibility of the Head/Principal. Ensuring stability and consistency. Is a very controlled and well-structured environment. Has lots of formal procedures which staff must follow.	Primarily focussed on developing and promoting a school vision. Distinctly separated from day to day management issues. Mentoring, coaching and staff development. Promoting and defending core educational values.
Participative	Predominantly about staff communication; guidance and feedback. Organised around staff participation and shared decision making. Mentoring, coaching and staff development. Strongly influenced/constrained by local community challenges. Supports teamwork, consensus, and participation. Provides excellent mentoring and support. Values the provision of new resources and creating new challenges. Fosters trust, openness and participation. Encourages staff to be creative and try out new things, even if this is outside of accepted norms.	Mostly involved with preserving the school culture and ethos and representing the figure of authority. Firmly the responsibility of the head/principal Major decisions about curriculum development. Building school vision and mission

Annex 8 – Analysis of Vision Variables Relative to Leadership Paradigms

Refer to paragraph 5.7.1.

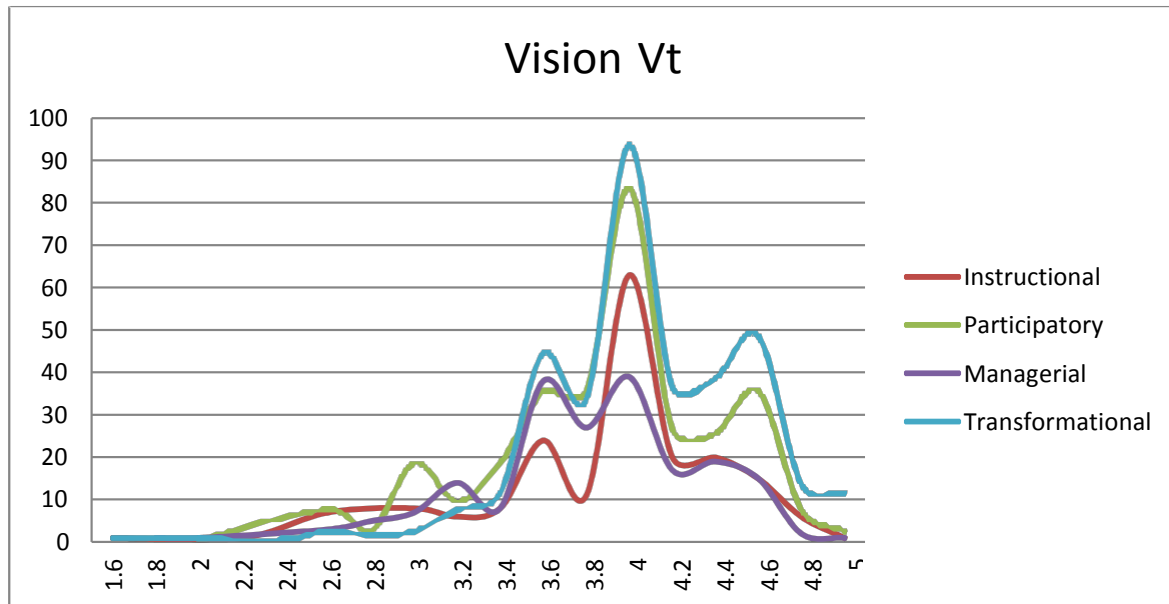


Figure A8.1 – Technology Vision relative to Leadership Paradigms

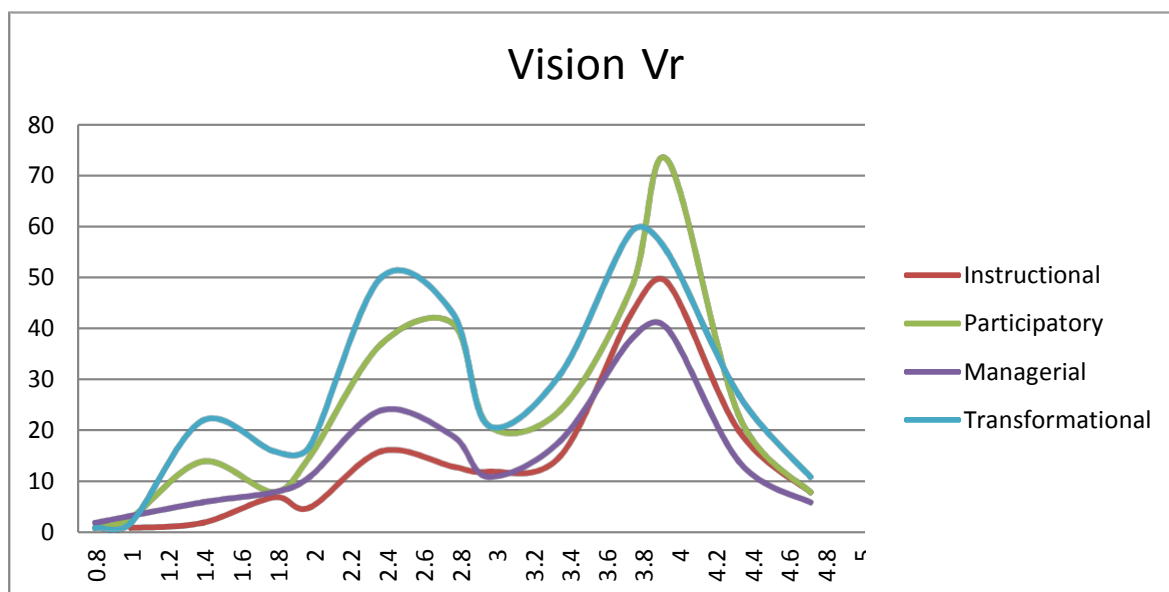


Figure A8.2 - Vision Responsibility relative to Leadership Paradigms

Annex 9 – Relationships between Intermediate and Dependent Variables

Refer to paragraph 5.7.3.

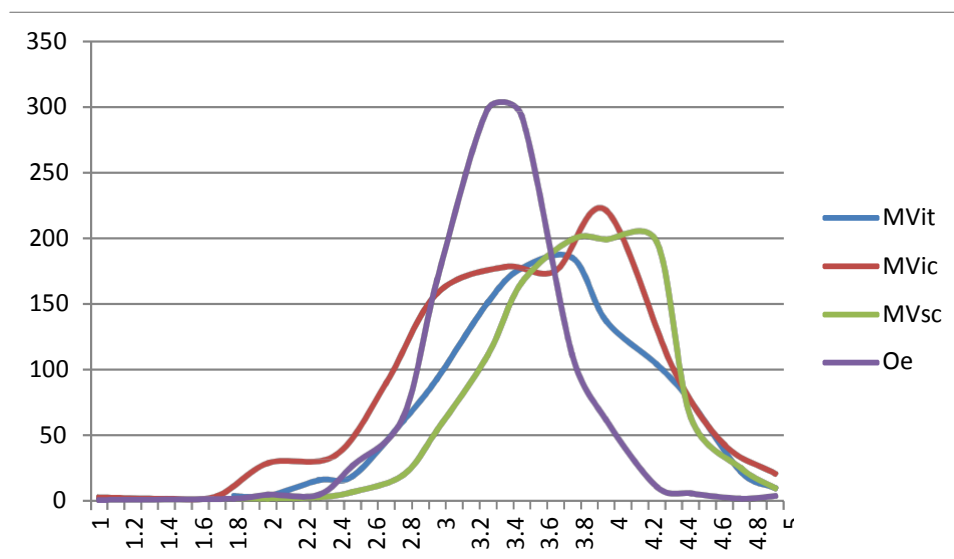


Figure A9.1 - Intermediate variable correlations with Educational benefits

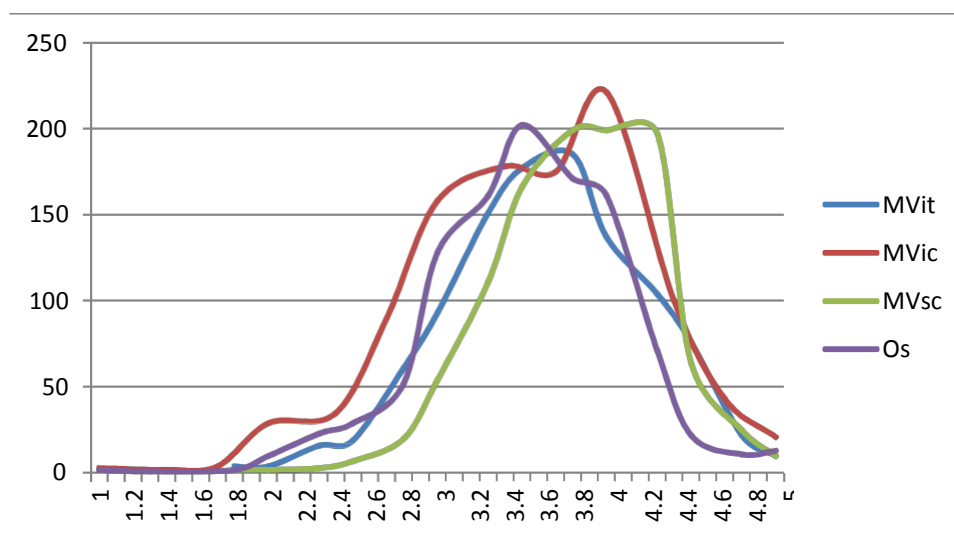


Figure A9.2 - Intermediate variable correlations with School benefits

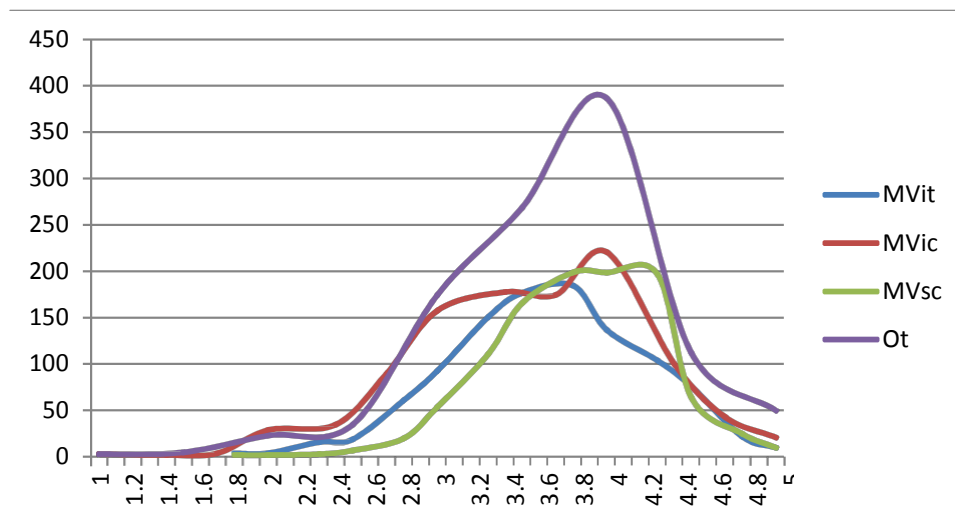


Figure A9.3 - Intermediate variable correlations with Teacher benefits

Annex 10 – Relationships between Independent and Intermediate Variables

Refer to paragraph 5.7.4.

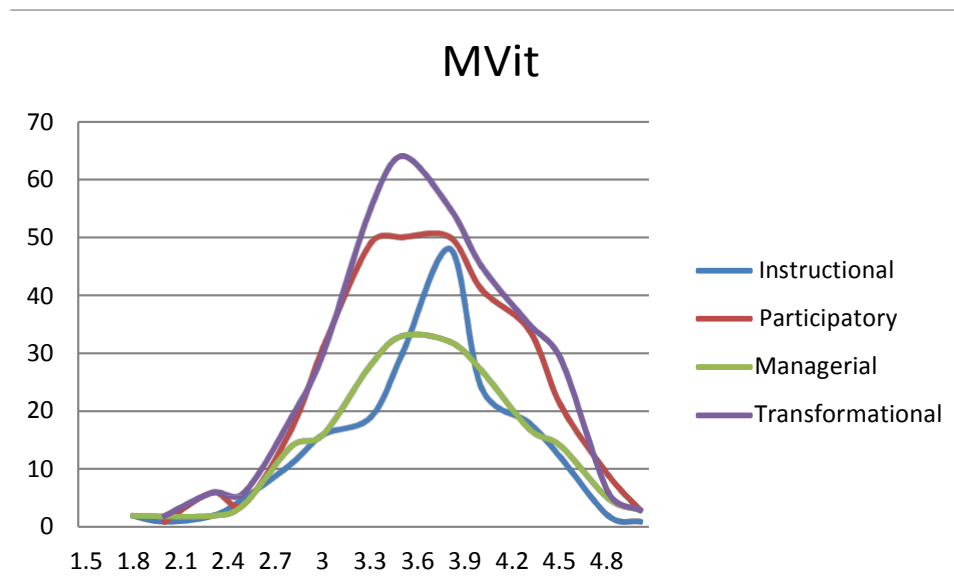


Figure A10.1 – Technology intermediate variable correlations with Leadership

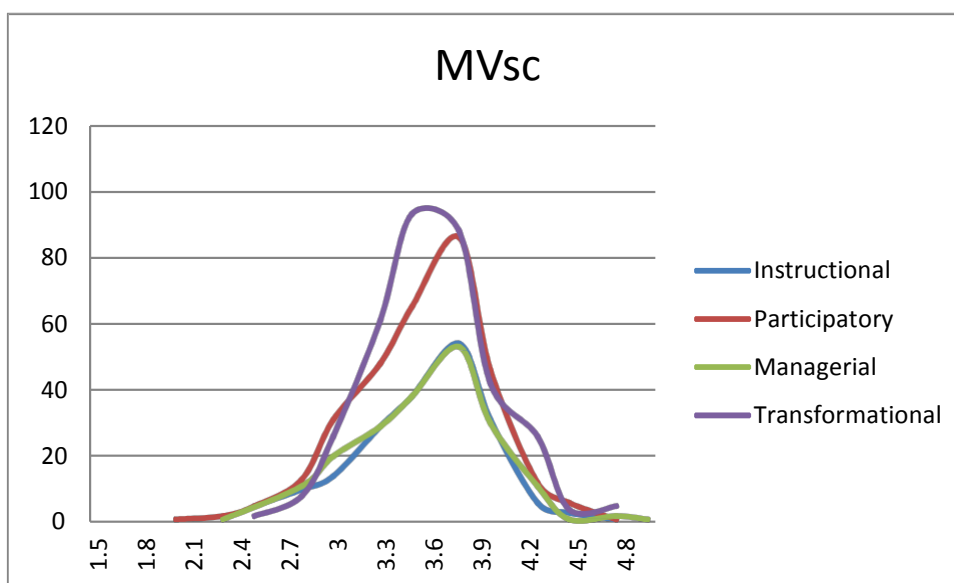


Figure A10.2 – School Culture intermediate variable correlations with Leadership

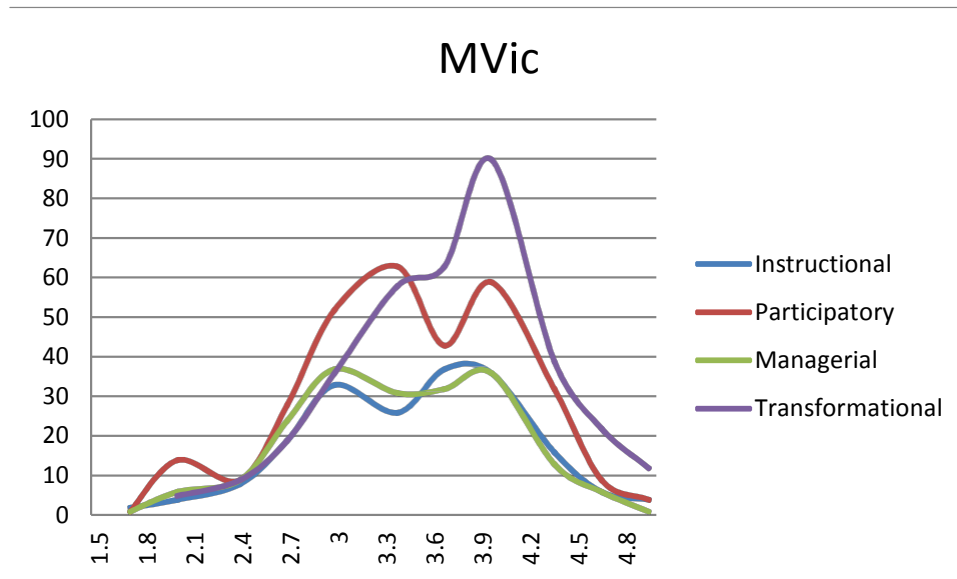


Figure A10.3 – Individual (Teacher) Culture intermediate variable correlations with Leadership

Annex 11 – Final reflections

These notes were written by Peter when we asked his permission to use his dissertation as an exemplar. They provide some more reflection on the process, several months after the dissertation was submitted and assessed.

The objective of this short paper is to present personal reflections on certain aspects of the dissertation process, and to consider important learning points.

In no particular order:

1. I really should have thought more carefully about the Residency advice to limit the scope of the project, even if this meant choosing a different topic. My project (involving a survey to all 22,000 UK schools, yielding 1600 responses), was way too ambitious. The amount of complex data arising from the survey resulted in an unbalanced dissertation process (in terms of demands on time and effort), and quite possibly, an unbalanced dissertation paper.
2. I should have resisted the temptation to start the research too early in the process. I thought I had a very clear understanding of the objectives of my research, however, when I really got into the depth of the literature review (from which I learnt a huge amount on the topic), I discovered numerous aspects that I really wished I'd approached differently in my research. Unfortunately, the survey (which was phased over several weeks) was already in progress, and couldn't be amended mid-way through.
3. I allowed the structure of my dissertation to get out of step with the marking scheme. The literature review was too comprehensive, my research too extensive, and my analysis too detailed. Even though I culled large amounts of each (discarding large sections from a dissertation that had swelled to well over 20,000 words), I did this in an unbalanced way. Each of the above sections was still too big in the final version, leaving insufficient space to properly cover Recommendations and other important areas. I'm sure I'll have lost marks as a result.
4. I put a lot of effort (over several weeks) into the statistical analysis of my survey data. Whilst this did improve the quality of the dissertation as a stand-alone piece of research, I feel sure this substantial effort wasn't *proportionately* reflected in the marking scheme. The time spent in studying statistical techniques, finding and learning appropriate software, manipulating and checking (and *re-checking*) data in Excel, could have been used more effectively on other things (eg, with my family!!).

5. The literature research got out of hand, mainly due to the relative ease with which research papers, books and e-books can now be searched and sourced using the internet. The bibliography of each successive paper often identified a further 10 or 20 potentially interesting and useful new sources, each of which could usually be located, downloaded and read. Each one had its own interesting bibliography, yielding yet more sources. The process of searching, downloading and reading became compulsive and very time consuming. With hindsight, I would establish a much more clearly defined regime for the literature research/review, and adopt a more aggressively selective process.
6. I had so much data (on so many aspects of my chosen topic) that my Research Findings section was too wide ranging and perhaps a little incoherent. At the time of writing, my intimate knowledge of each topic meant that the superficial summary I gave for each made perfect sense – but only because I had a detailed understanding of what lay beneath. To the markers (or any other ‘casual’ reader – including myself, a couple of months later), there’s simply too much information, too many interrelated topics, and too little depth. (This is redressed somewhat towards the end of the Findings section and in the Conclusions, but even so, too many of the early pages contain material on a range of interrelated topics, that is too superficial and difficult for the reader to integrate into a cohesive narrative).
7. I don’t think I fully benefitted from my supervisor during the process. Being unsure of the quantity (and nature) of involvement that was ‘allowed’, I very much tended to go it alone, with very infrequent requests for input. I now know that other MALIC students had a much closer relationship with their supervisors; with hindsight, I’d probably have been more demanding, or at least, more communicative.

Finally, to any MA student, my dissertation advice would be to reflect on the question “*why are you doing this, and how are you hoping to benefit*”? For many, I’m sure it’s mainly about the qualification; for some, it’s genuinely about the personal development; for a few, the research topic itself will be a significant interest and attraction.

The most effective dissertation strategy is intimately related to these motivations, affecting multiple aspects of planning and execution; and also, of course, how each of us will judge the outcomes.